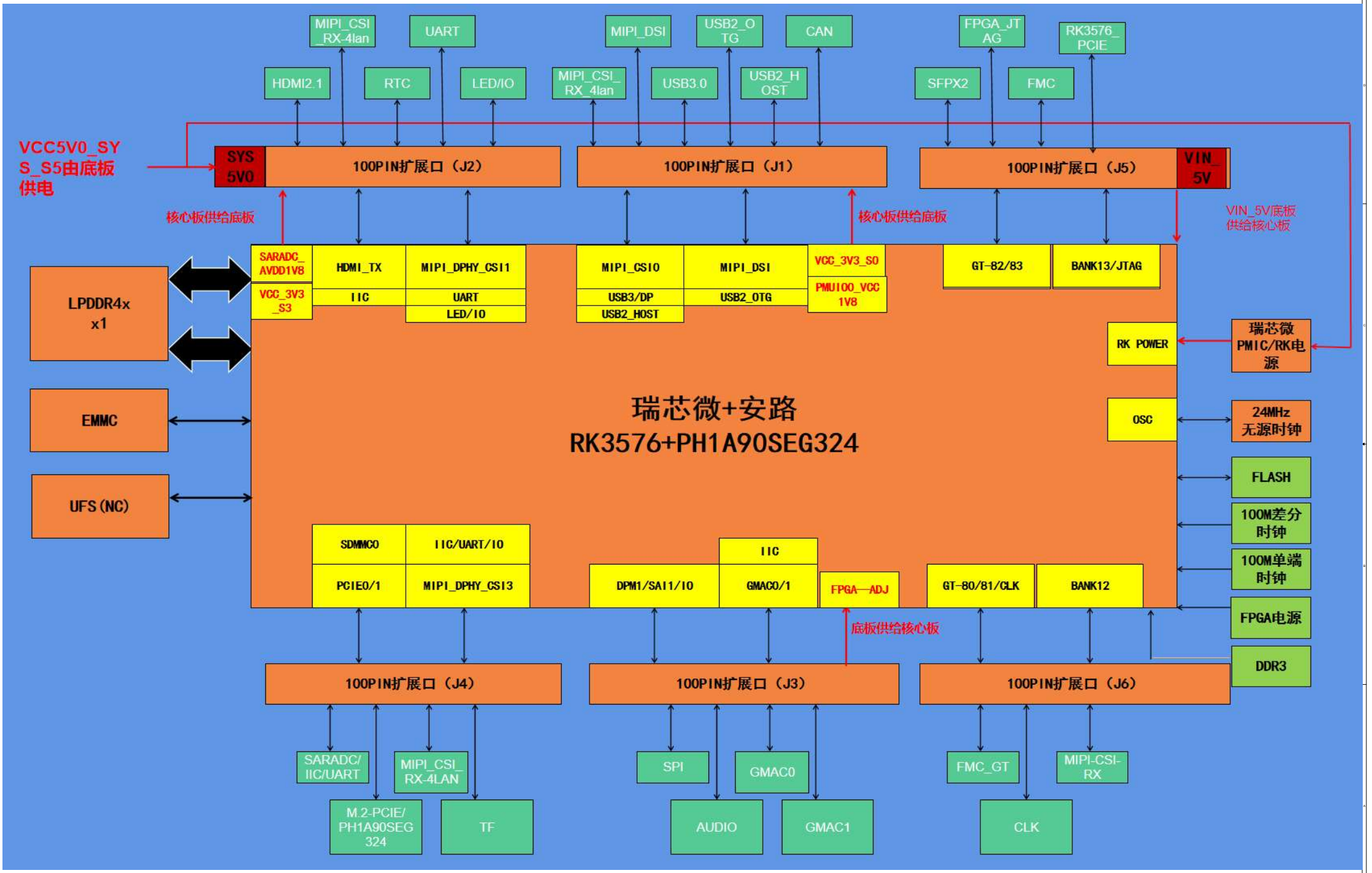
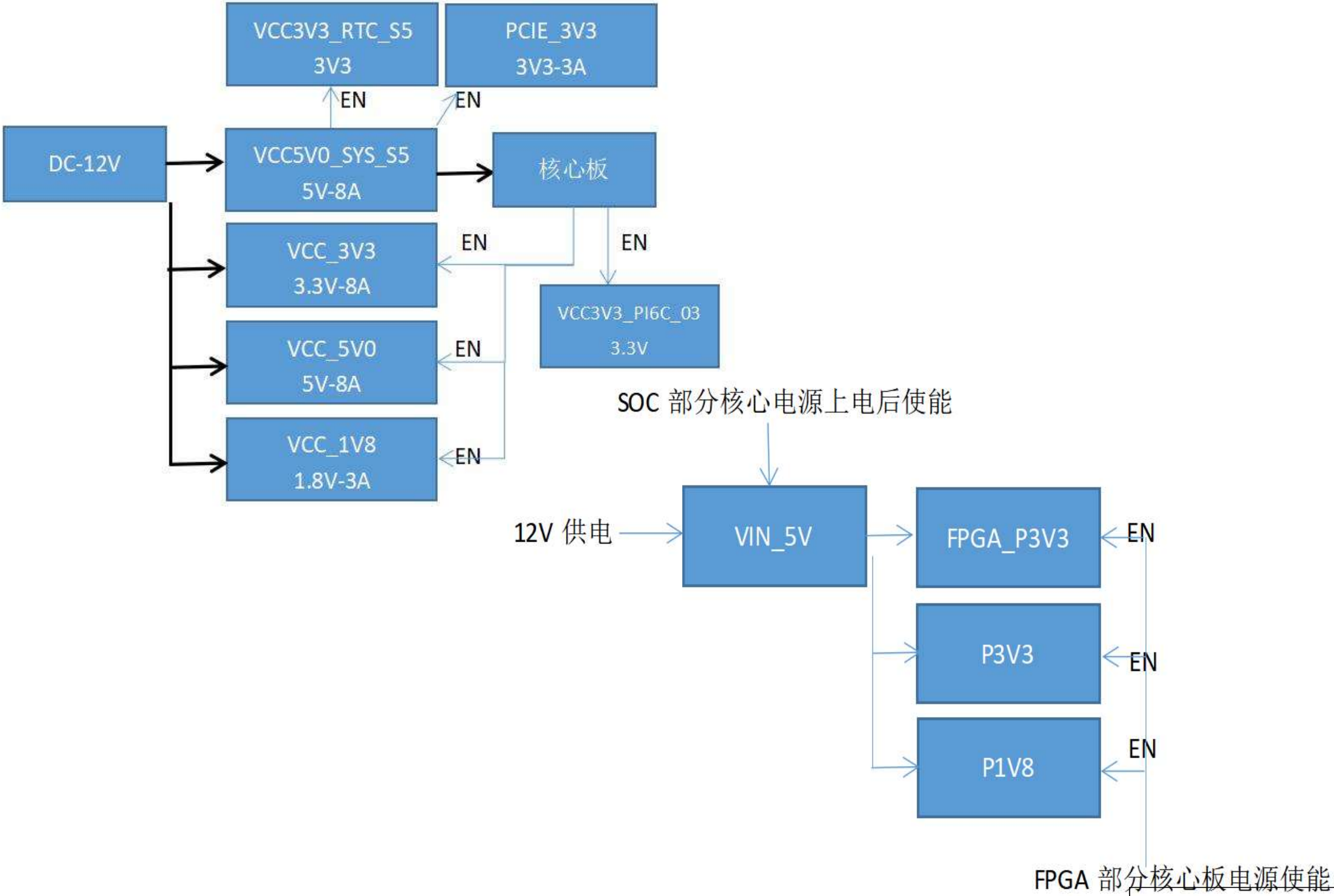
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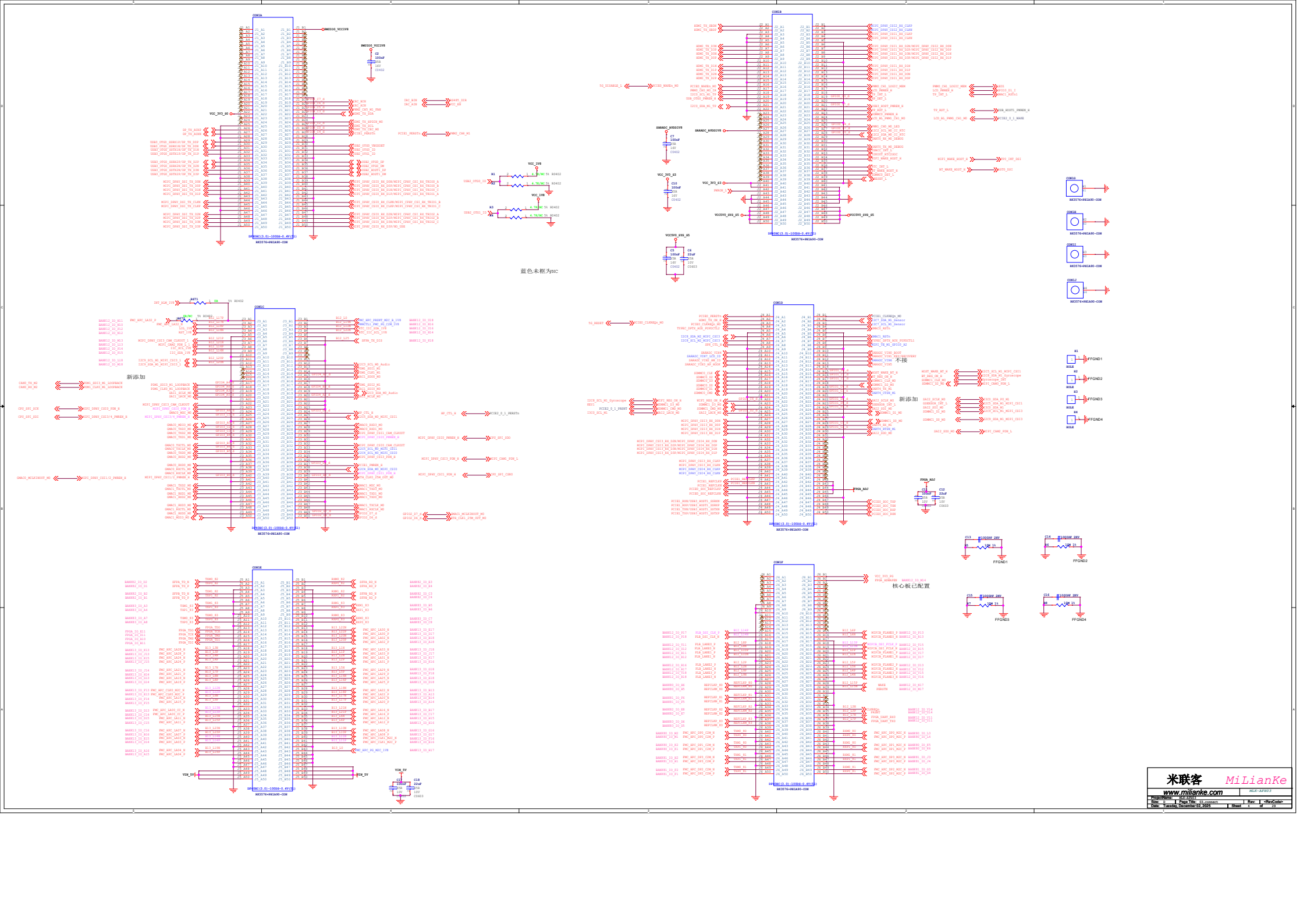
Page 1	00.Contents
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Page 3	02.Power Sequence
Page 4	03.connect
Page 5	04.Power_RTC_FAN
Page 6	05.DebugUART
Page 7	06.USB/TypeC Port
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Page 26	25.FPGA_EP_RTC
Page 27	26.FPGA_RK3576_PCIE_RC_EP
Page 28	27.FPGA_SFP
Page 29	28.FPGA_FMC

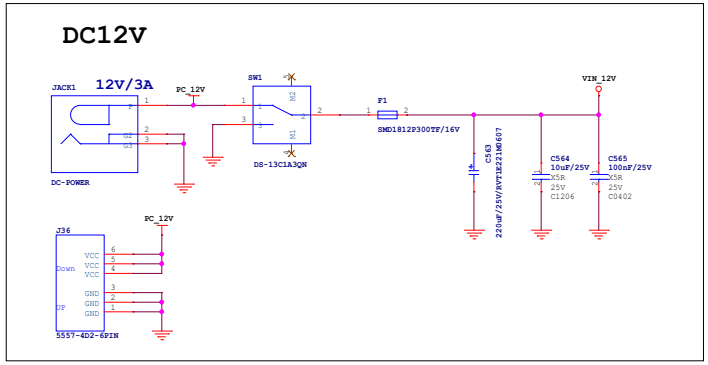


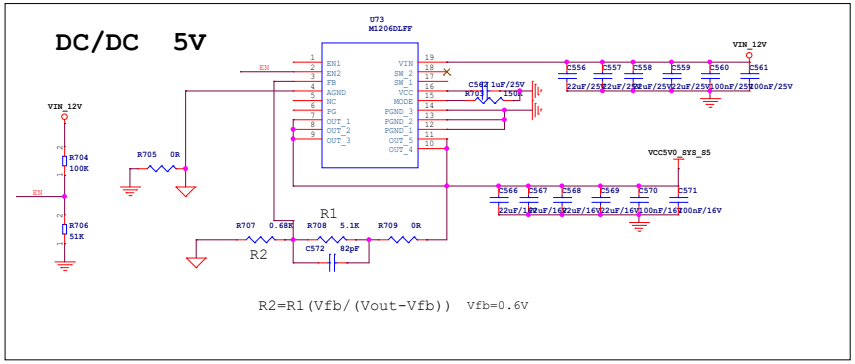
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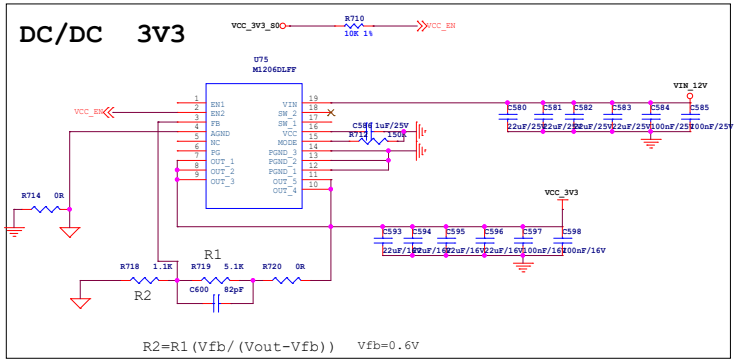


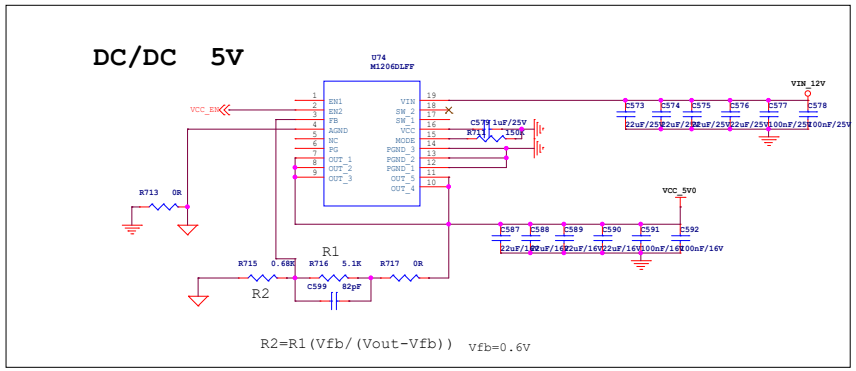
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MLK-AFH03
ProjectName: MLK-AFH03
Size: A2 | Page Title: 02_Power Sequence | Rev: <RevCode>
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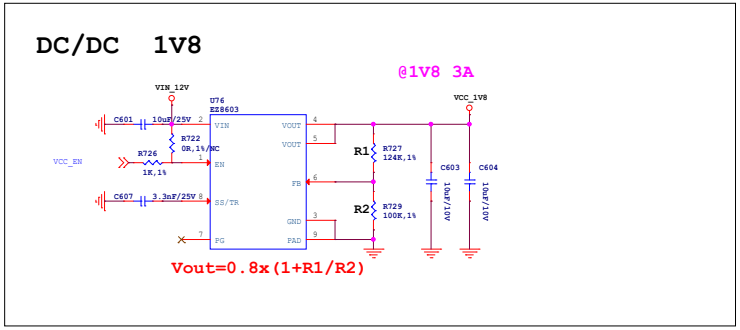

蓝色未框为NC
新添加
新添加
不接
核心板已配置
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MLK-AF103
Project Name: MLK-AF103
Size: D
Page Title: 03_connect
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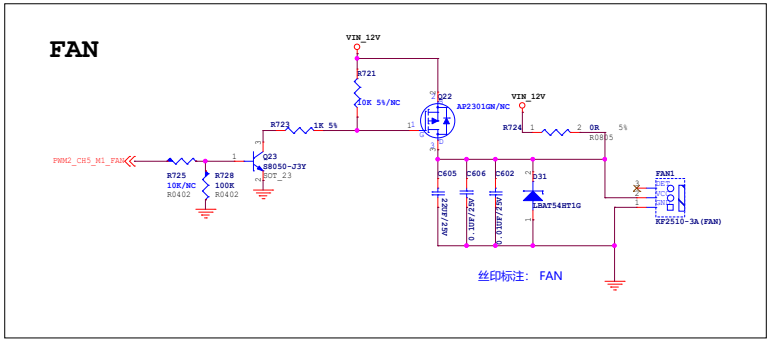


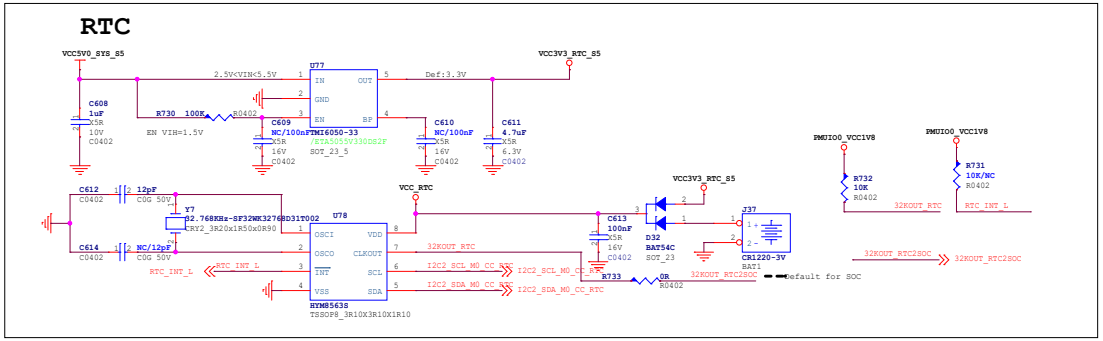






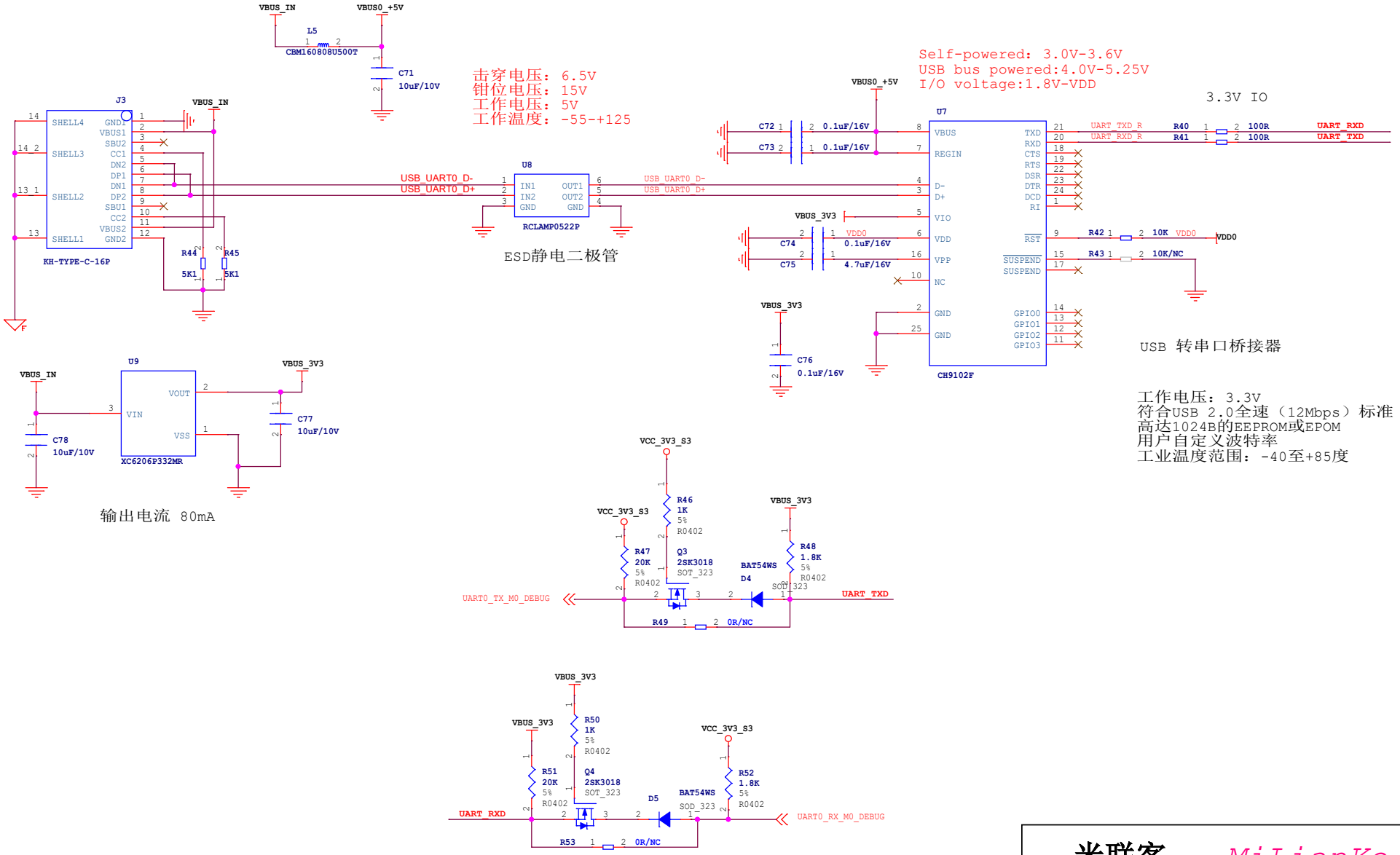






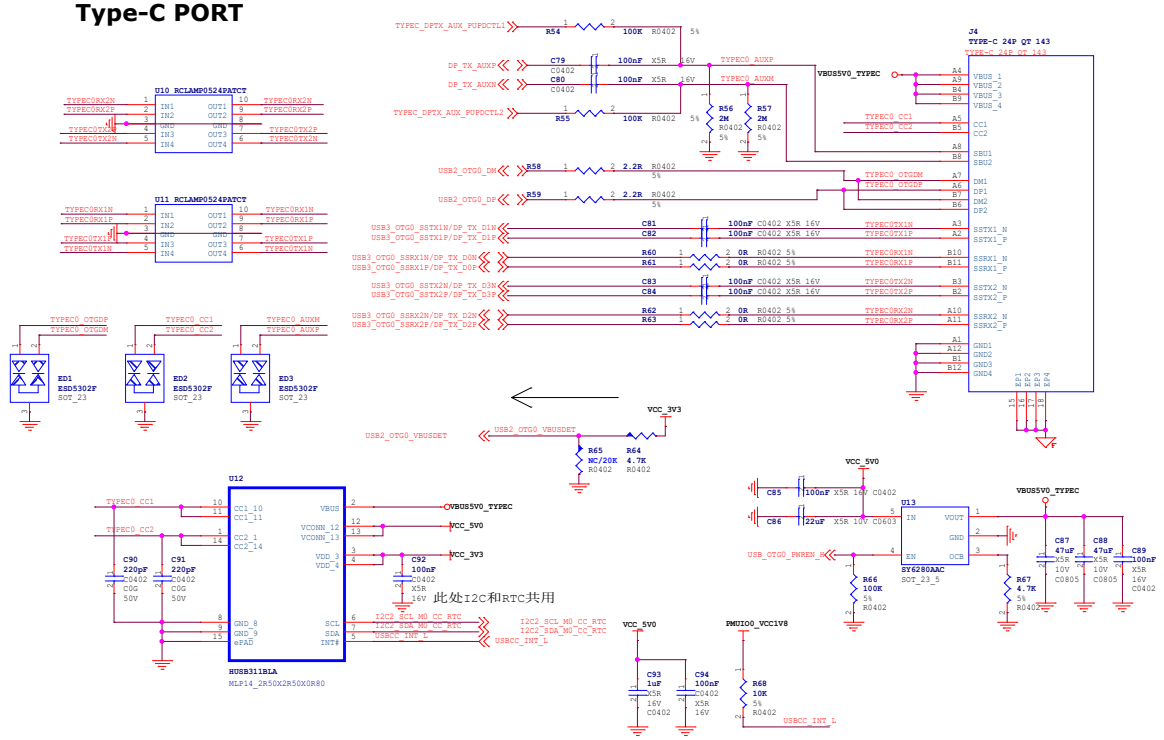
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UART (DEBUG)

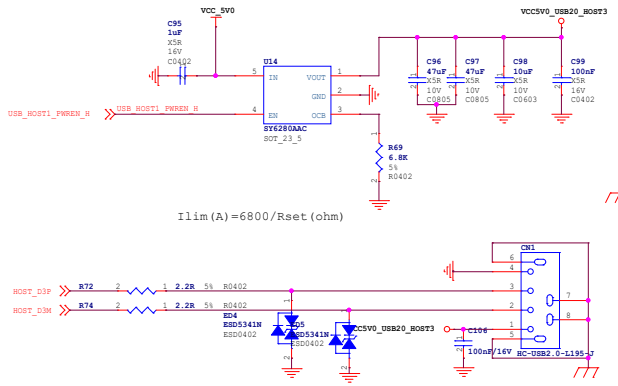


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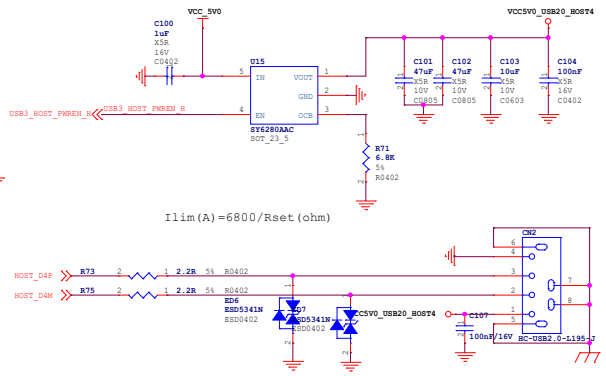
Type-C PORT



USB2.0_CN1



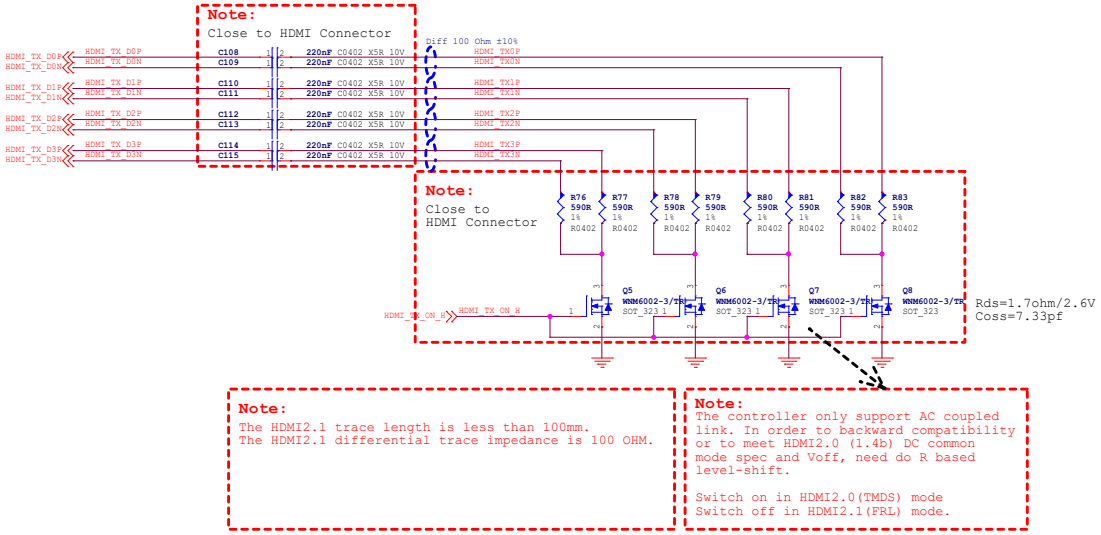
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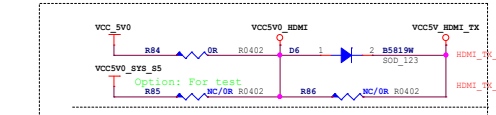


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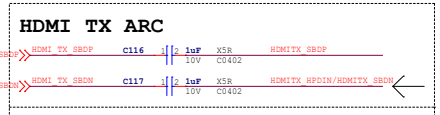
HDMI 2.1-TX

Support video output up to 4Kx2K@120Hz

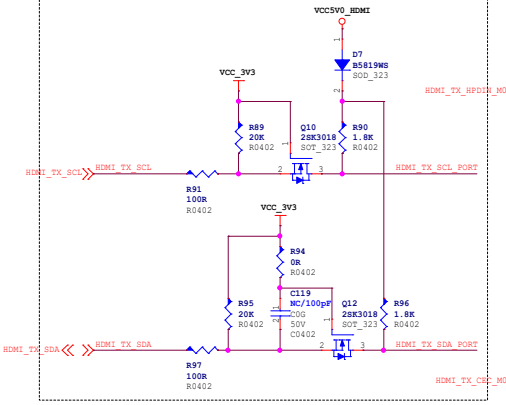




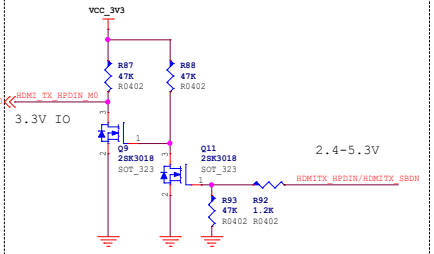
HDMI TX ARC



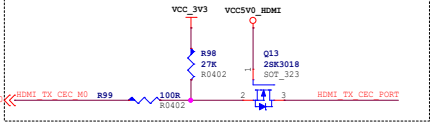
HDMI TX DDC



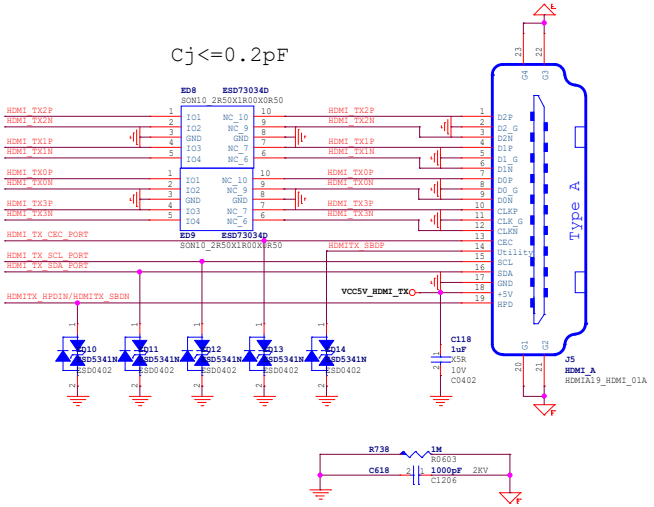
HDMI TX HPD



HDMI TX CEC



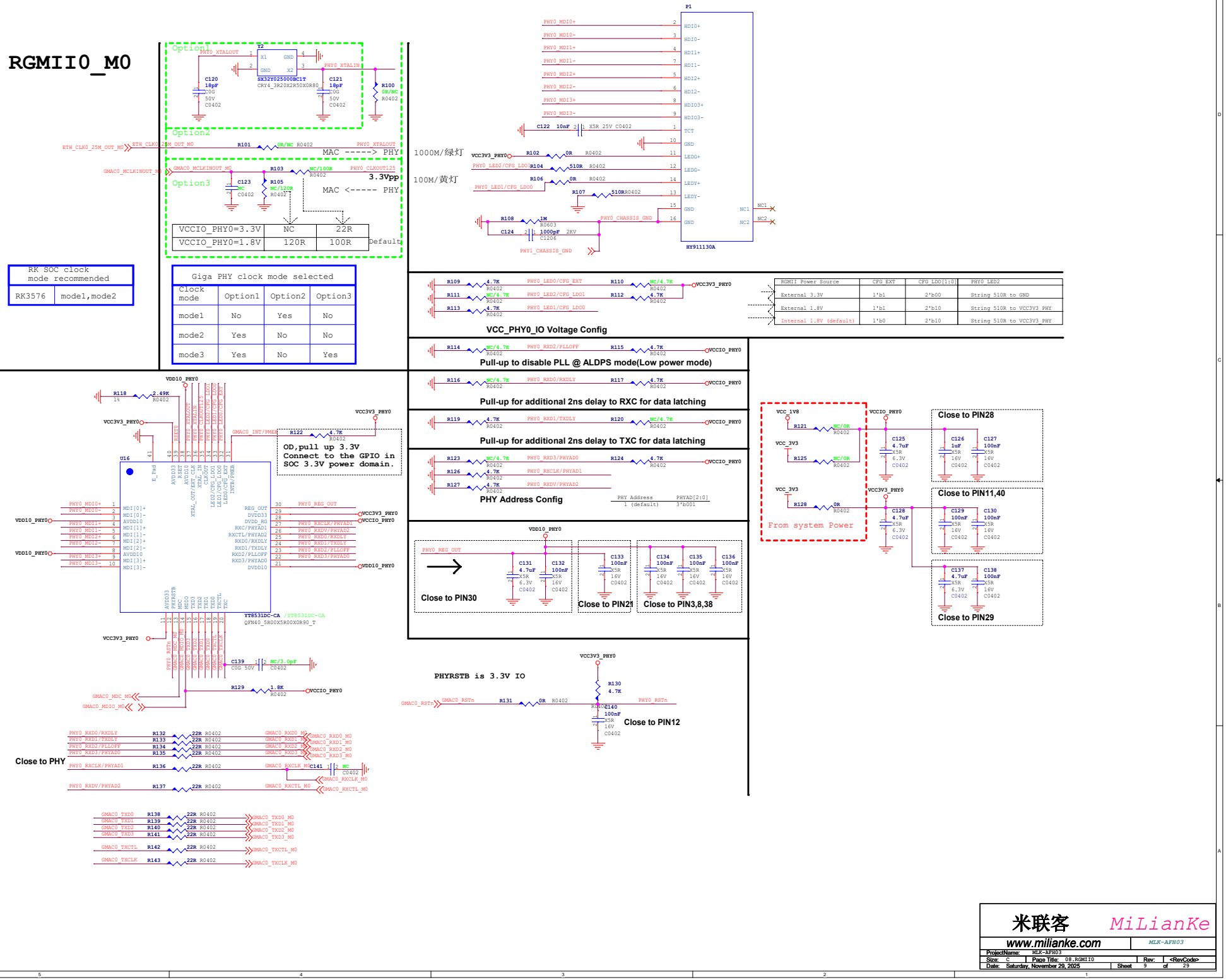
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RGMII0_M0

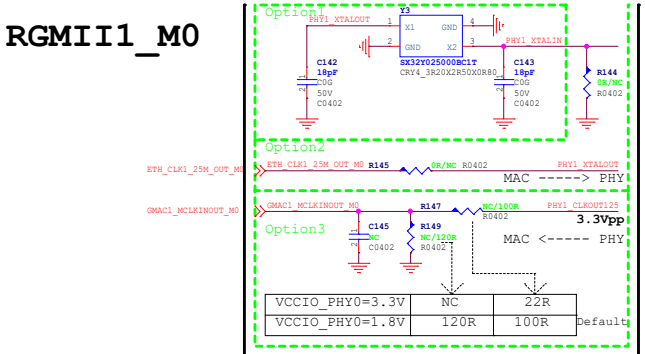


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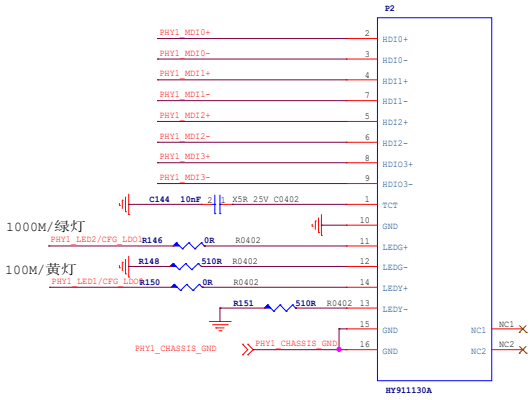
RGMII1_M0

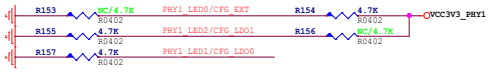


VCCIO_PHY0=3.3V	NC	22R	
VCCIO_PHY0=1.8V	120R	100R	Default

RK SOC clock mode recommended	
RK3576	mode1,mode2

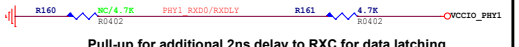
Giga PHY clock mode selected			
Clock mode	Option1	Option2	Option3
mode1	No	Yes	No
mode2	Yes	No	No
mode3	Yes	No	Yes

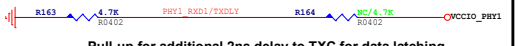


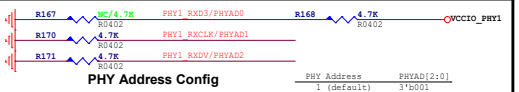


RGMII Power Source	CFG EXT	CFG LDO[1:0]	PHY1 LED2
External 3.3V (default)	1'b1	2'b00	String 510R to GND
External 1.8V	1'b1	2'b10	String 510R to VCC3V3_PHY
Internal 1.8V	1'b0	2'b10	String 510R to VCC3V3_PHY

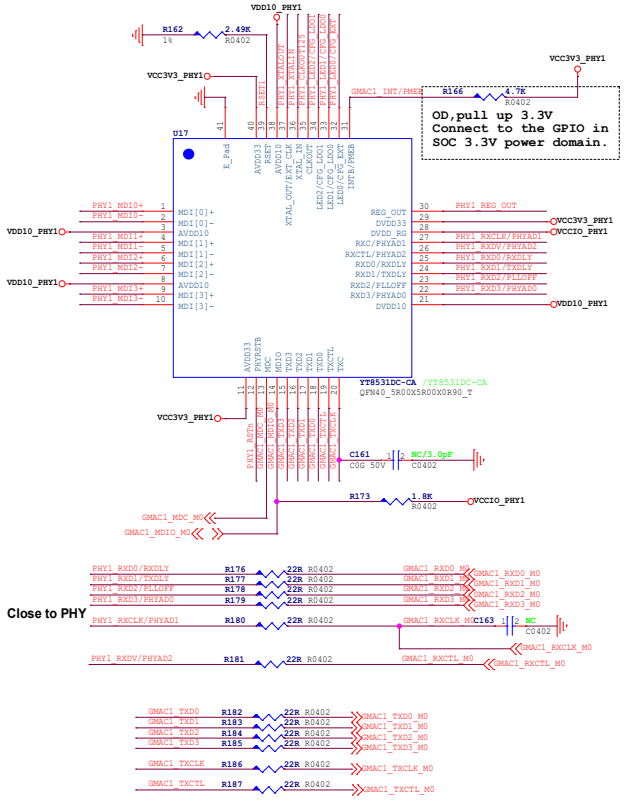


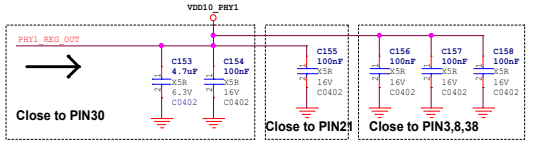


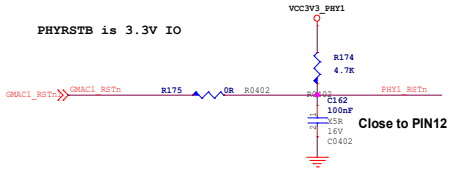


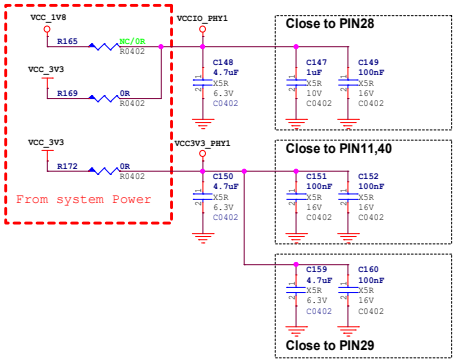


PHY Address	PHYAD[2:0]
1 (default)	3'b001



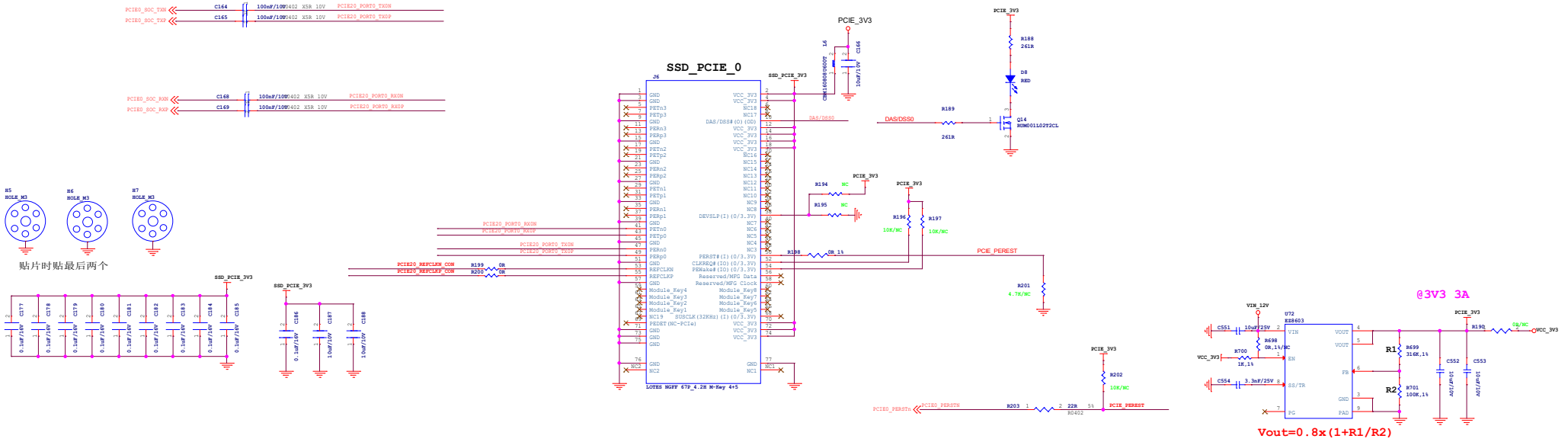


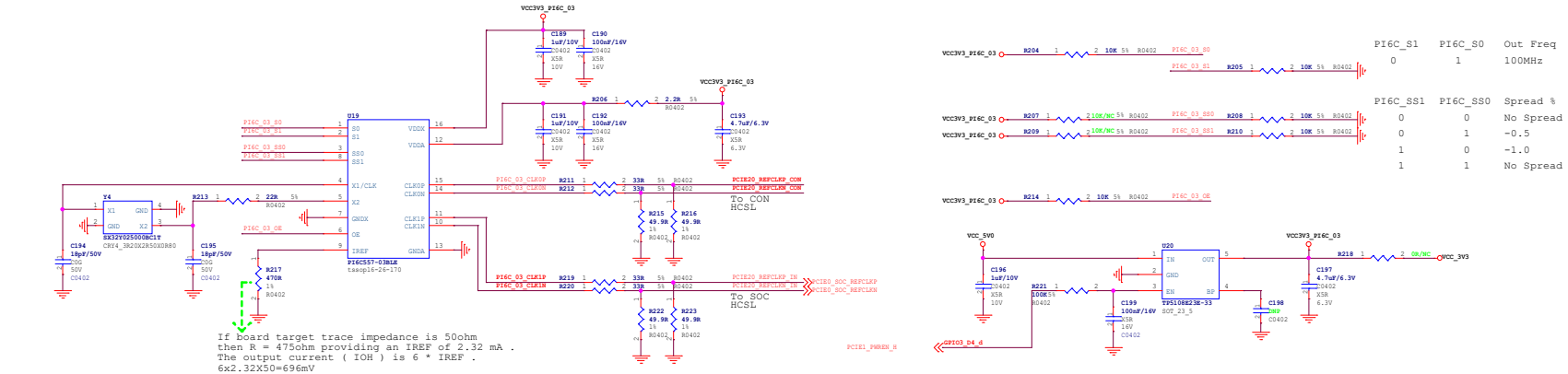




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M.2_PCIe2.0 x 1Lanes



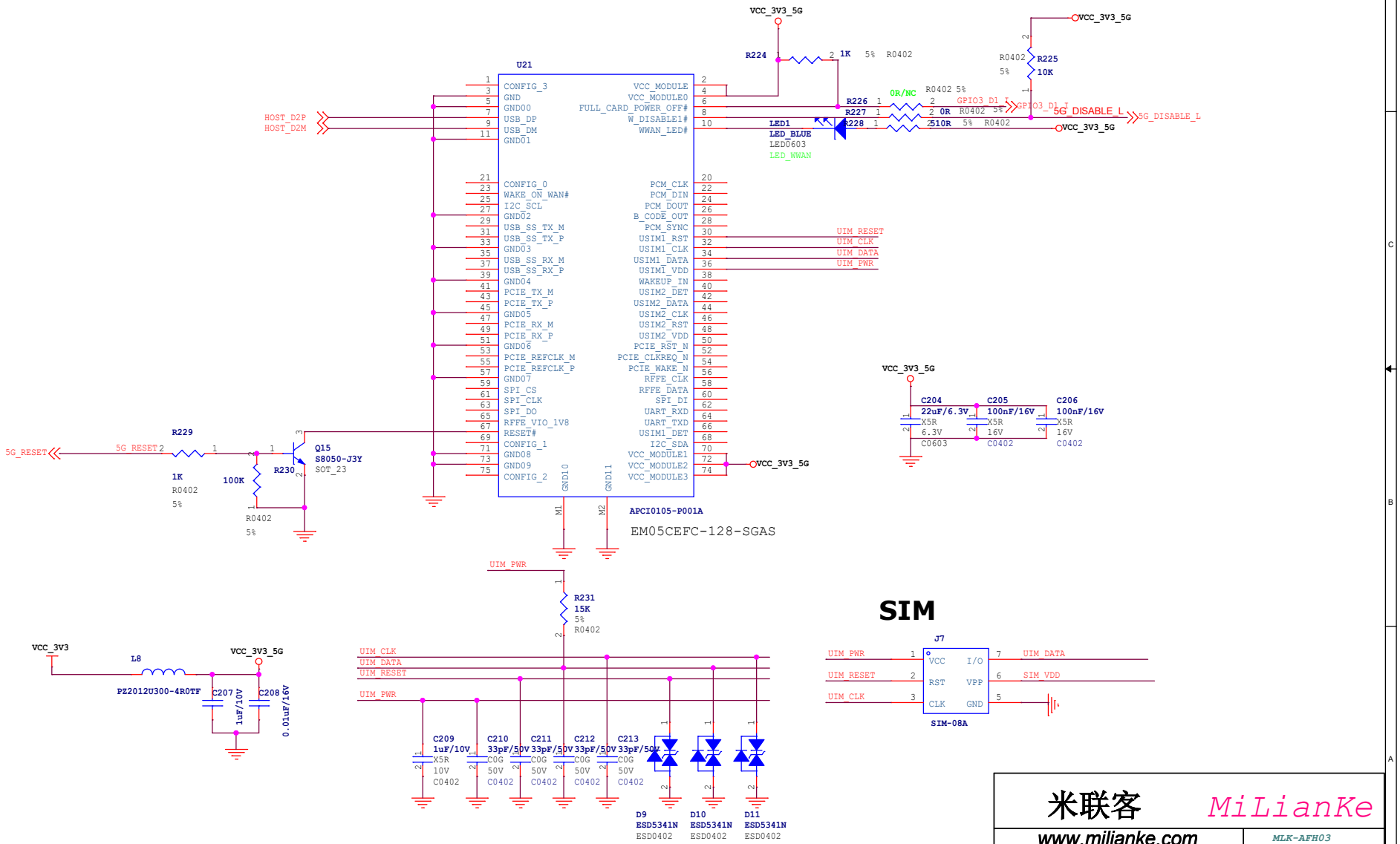


PI6C_S1	PI6C_S0	Out Freq
0	1	100MHz

PI6C_SS1	PI6C_SS0	Spread %
0	0	No Spread
0	1	-0.5
1	0	-1.0
1	1	No Spread

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MiniPCIe2.0 Slot_Support 4G module



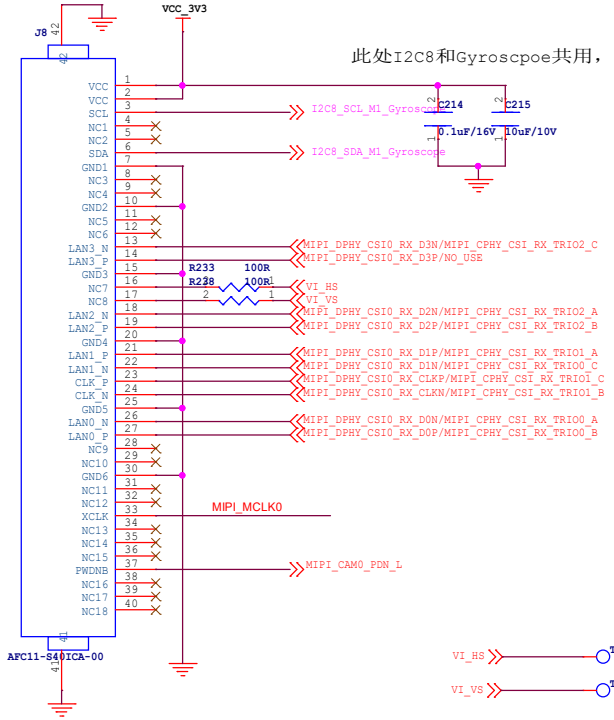
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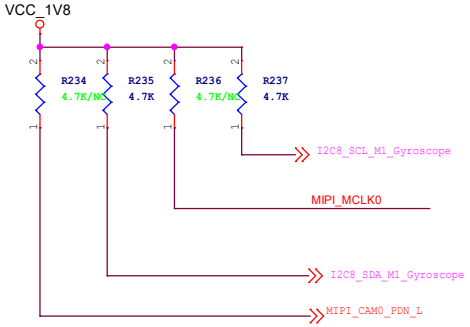
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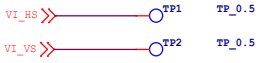
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VI-Camera CSI0_RX

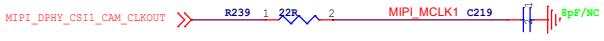


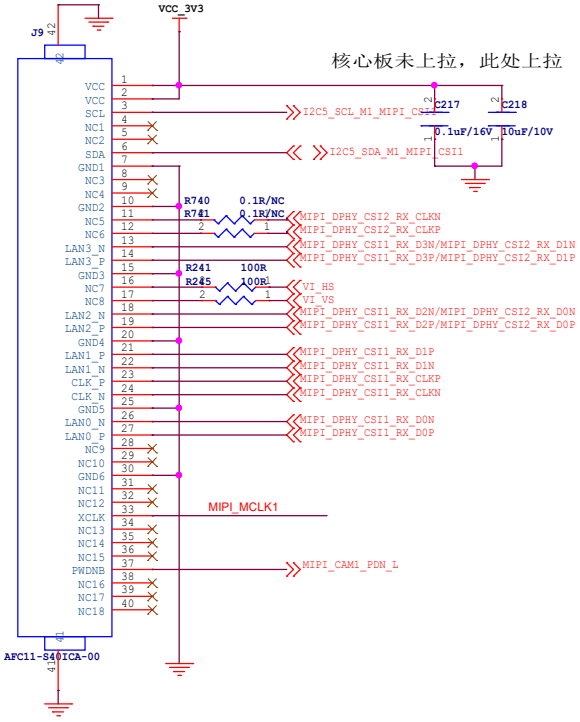


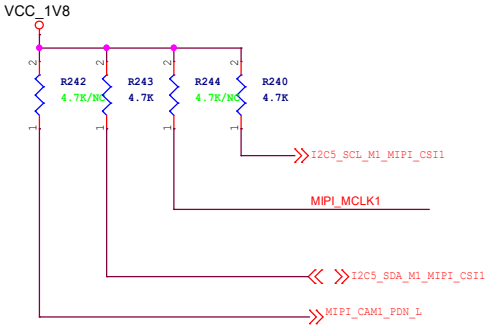




VI-Camera CSI1_RX





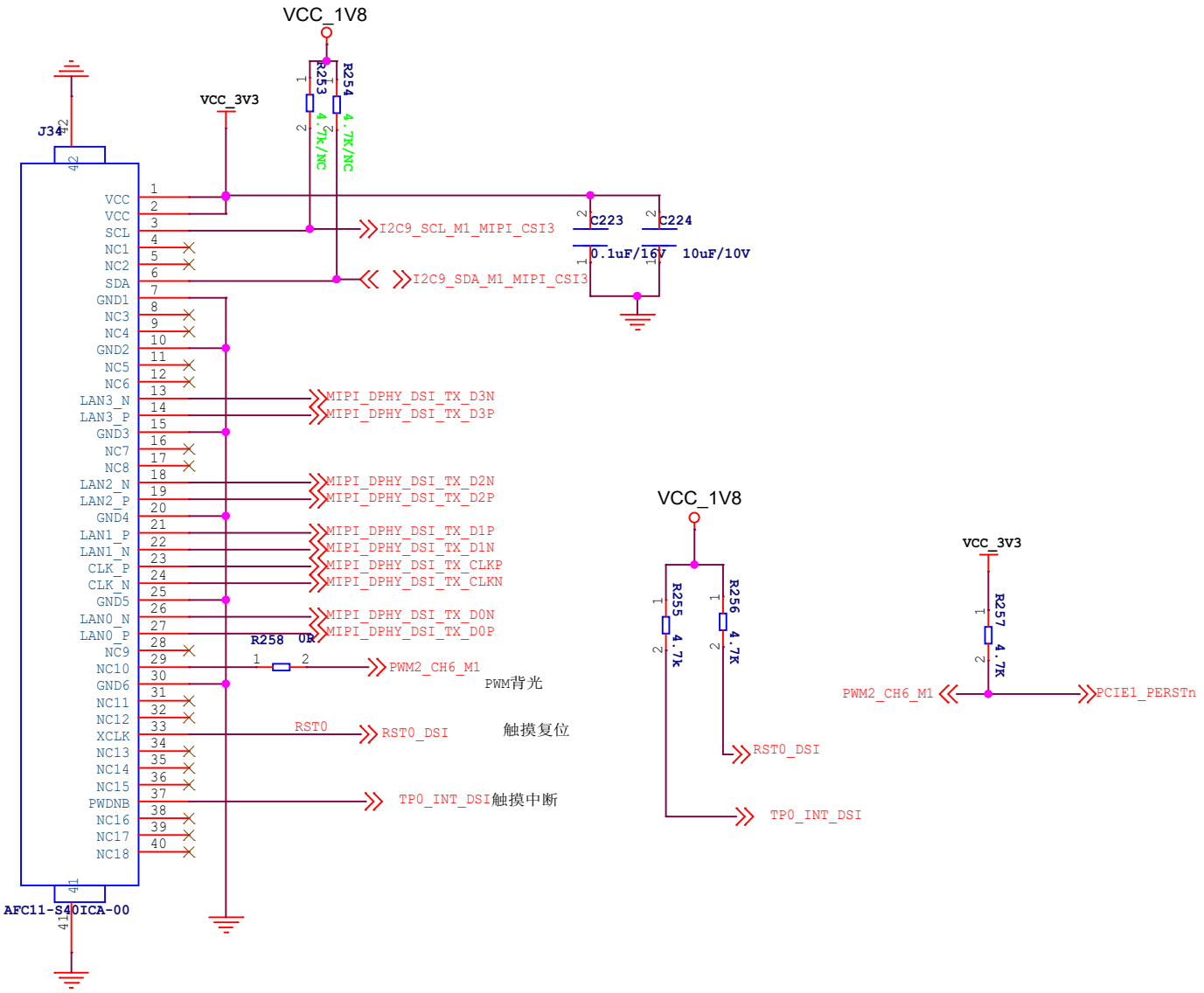


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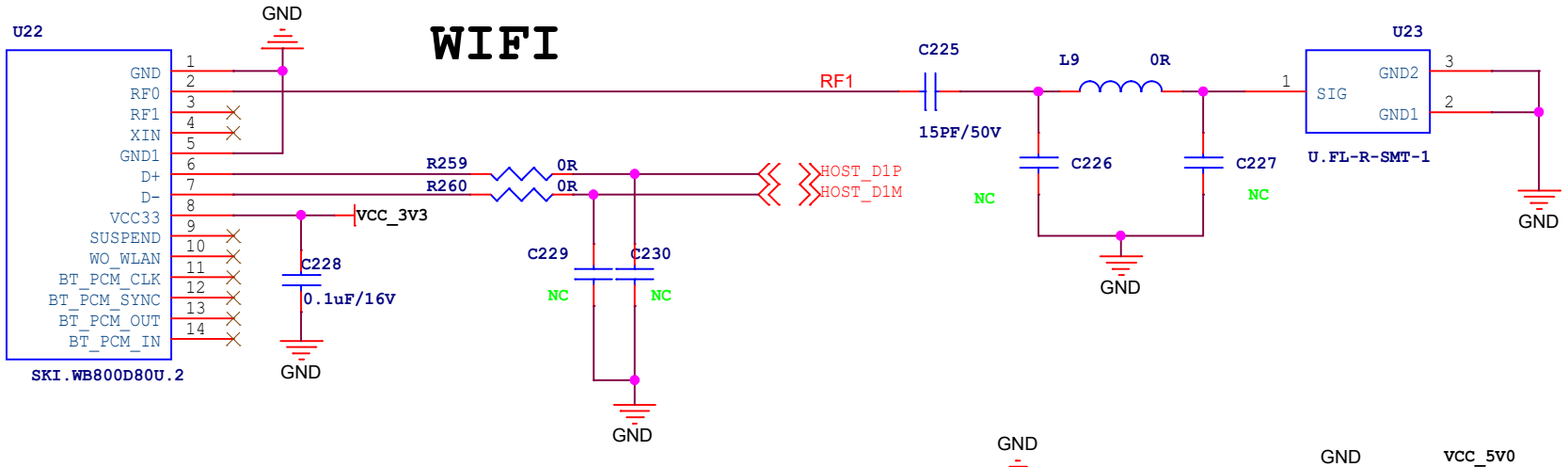
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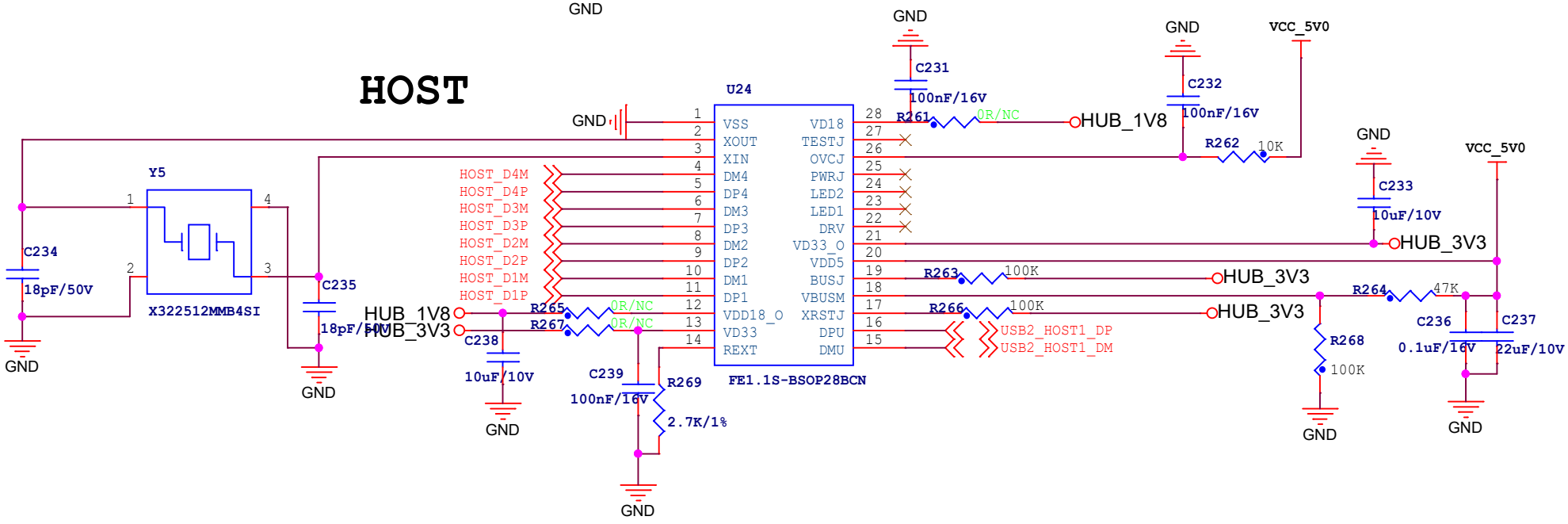
MIPI DSI TX



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PCIe WIFI/BT USB-HUB



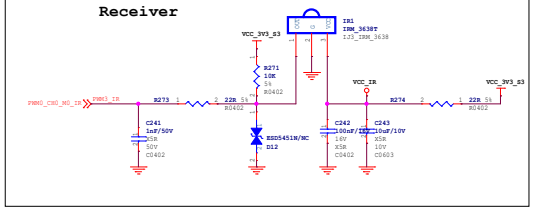


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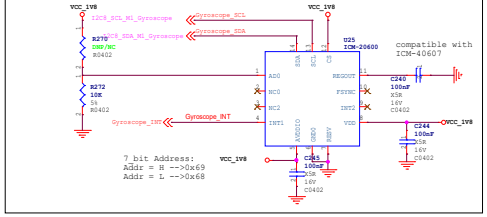
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Size: A	Page Title: 14.WIFI HOST	Rev:	<RevCode>
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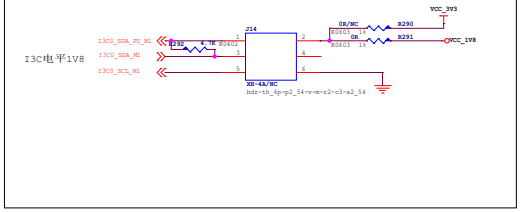
IR Receiver



Gyroscope+G-sensor

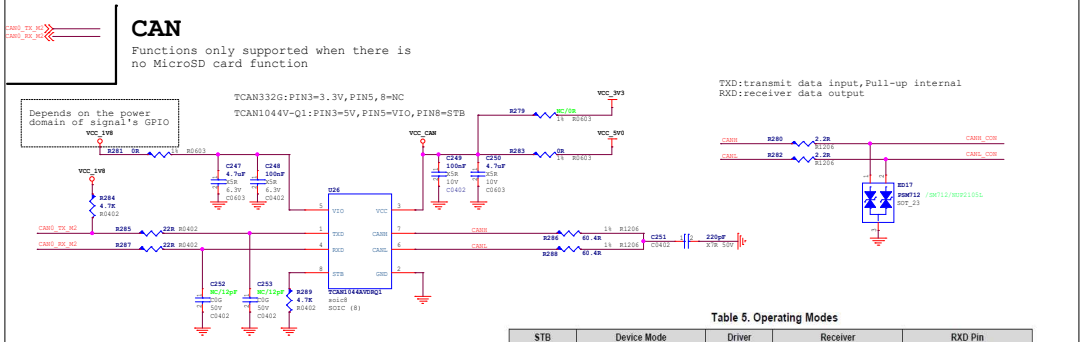


I3C



CAN

Functions only supported when there is no MicroSD card function



TXD:transmit data input,Pull-up internal
RXD:receiver data output

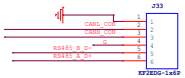


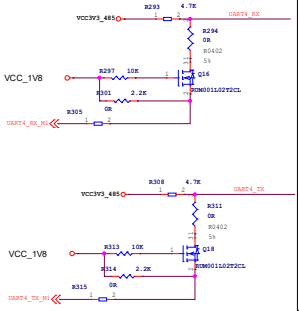
Table 5. Operating Modes

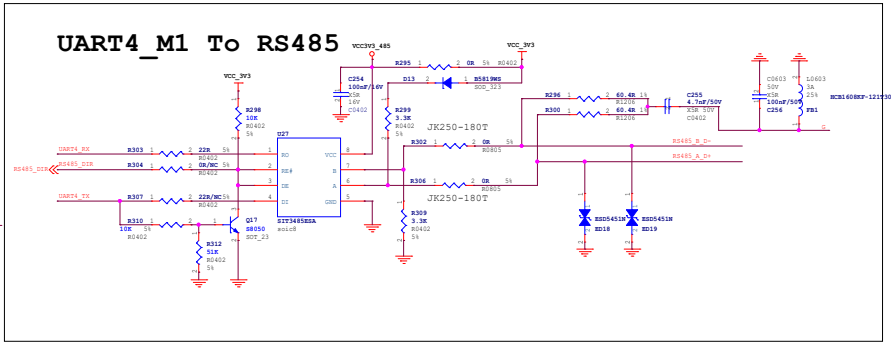
STB	Device Mode	Driver	Receiver	RXD Pin
High	Low current standby mode with bus wake-up	Disabled	Low-power receiver and bus monitor enable	High (recessive) until valid WUP is received See section 8.3.3.1
Low	Normal Mode	Enabled	Enabled	Mirrors bus state

TCAN1044V:
Vio:Support 1.8V,2.5V,3.3V,5V
Receiver common mode input voltage:+/-12V;
Support of classical CAN and optimized CAN FD performance at 2,5,and 8 Mbps.
Temperature grade:-40 to 125

STB(Standby)
The STB pin is an input pin used for mode control of the transcervier.The STB pin can be supplied from either the system processor or from a static system voltage source,If normal mode is the only intended mode of operation than the STB pin can be tired directly to GND.

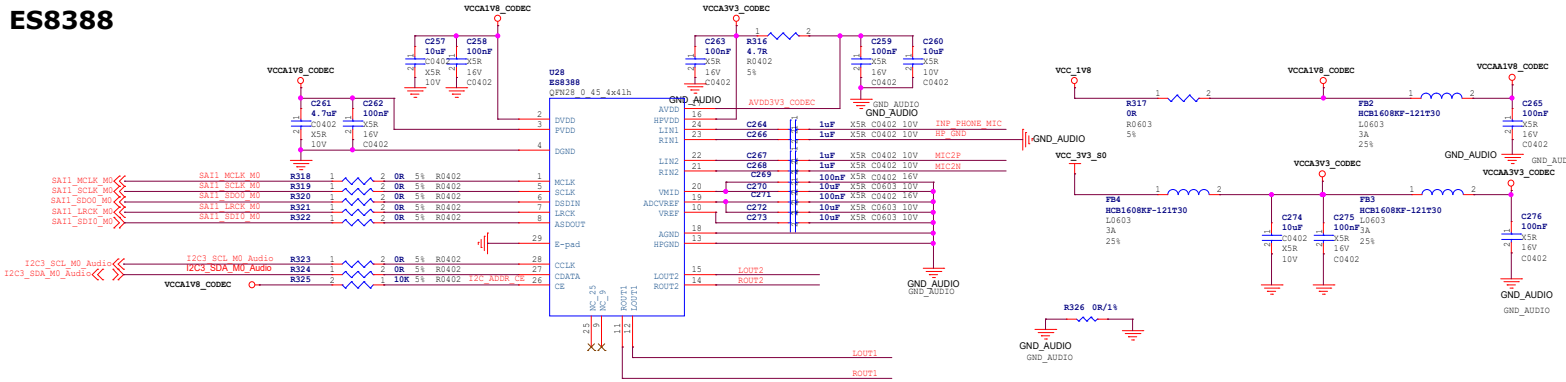
UART4_M1 To RS485



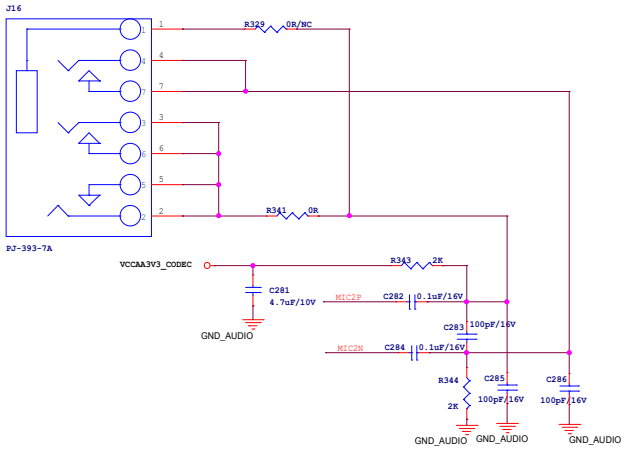


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ProjectName: MLK-AM03
Size: Custom | Page Title: 15.Sensors_485 | Rev: <RevCode>
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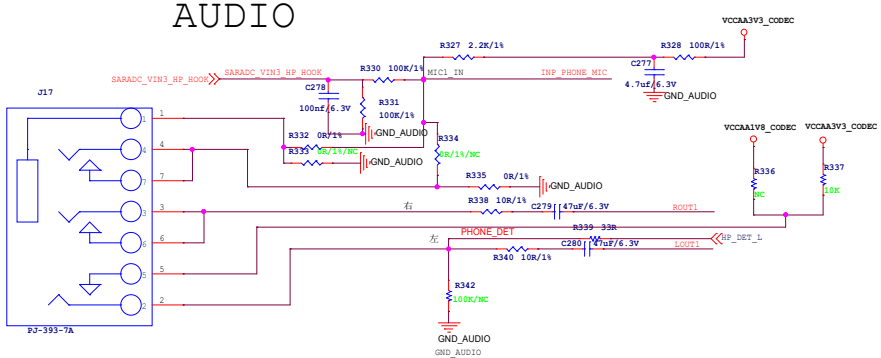
CODEC ES8388



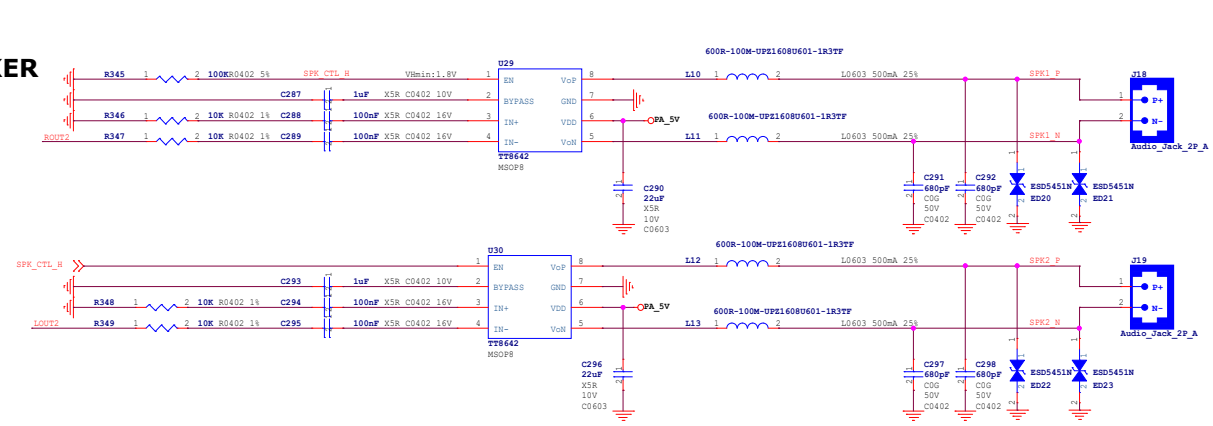
MIC



AUDIO



SPEAKER

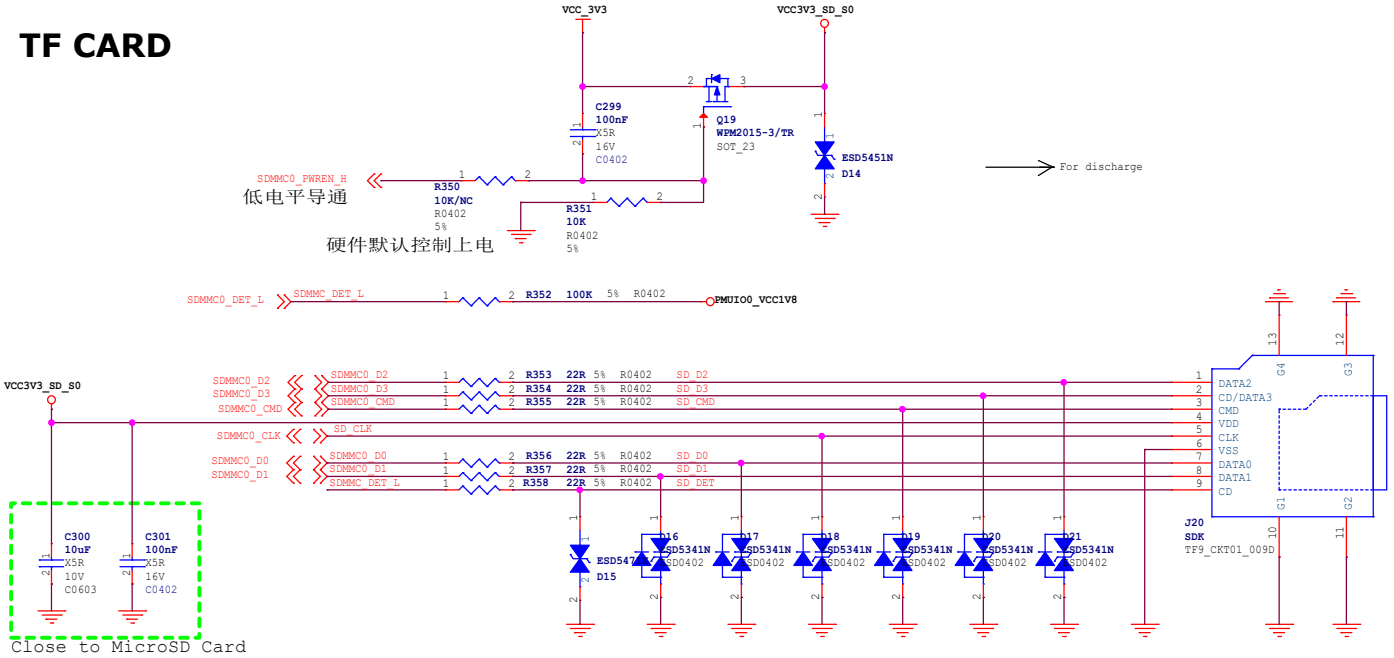


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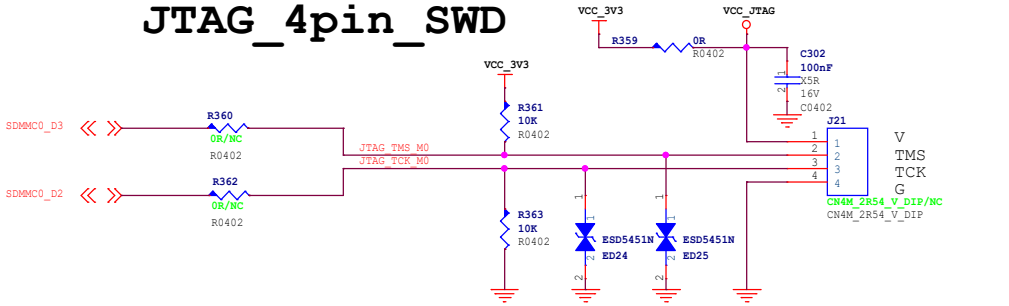
ProjectName:	MLK-AFH03
Size: C	Page Title: 16.MIC_PHONE
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	Rev: <RevCode>

TF CARD



MicroSD Card

JTAG_4pin_SWD

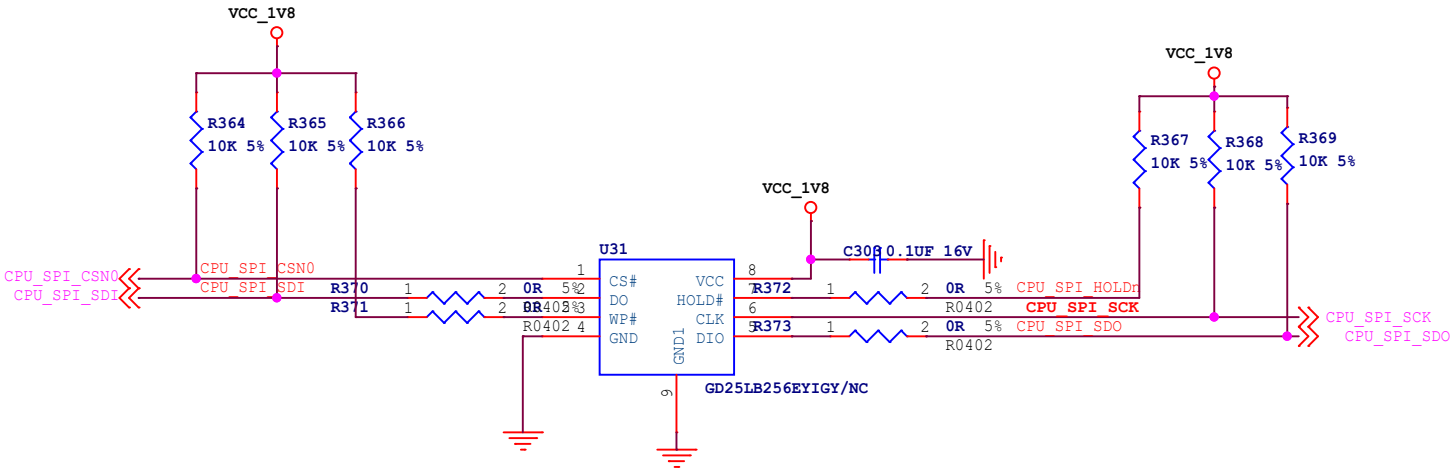


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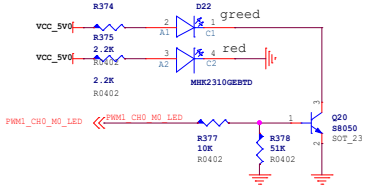
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Size: A3	Page Title: 17.SD Card	Rev:	<RevCode>
Date: Saturday, November 29, 2025	Sheet 18 of 29		

SPI Flash

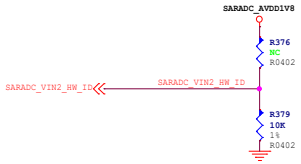


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ProjectName: MLK-AFH03			
Size: A4	Page Title: 18.flash	Rev:	<RevCode>
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LED

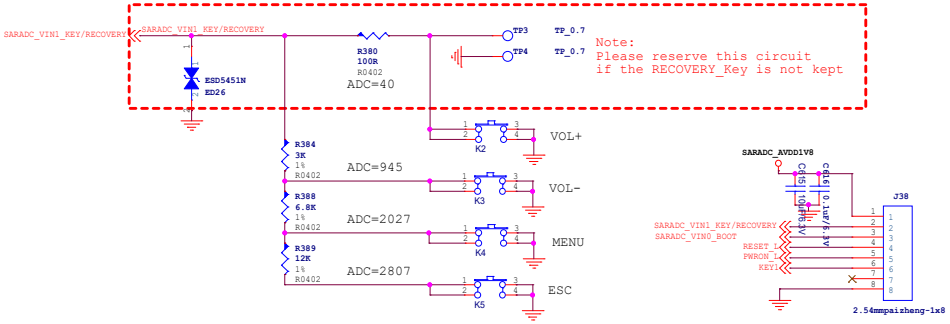


HW_ID

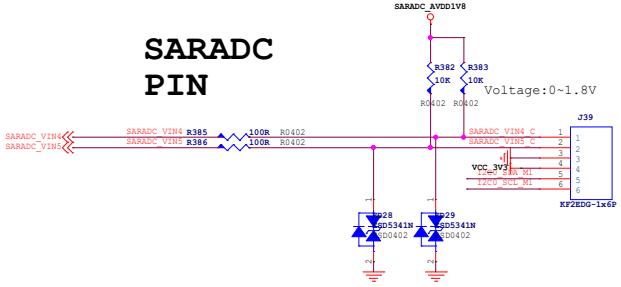


Config Table for SARADC_VIN2_HW_ID				
Item	Rup	Rdown	ADC Value	VERSION
HW_ID1	NC	10K	0	RK3576 EVB1 V12
HW_ID2	10K	1.13K	416	RESERVE
HW_ID3	10K	2.49K	816	RESERVE
HW_ID4	10K	4.3K	1231	RESERVE
HW_ID5	10K	6.8K	1658	RESERVE
HW_ID6	10K	10K	2048	RESERVE
HW_ID7	10K	14.7K	2437	RESERVE
HW_ID8	10K	23.2K	2862	RESERVE
HW_ID9	10K	40.2K	3279	RESERVE
HW_ID10	10K	88.7K	3680	RESERVE
HW_ID11	10K	NC	4095	RESERVE

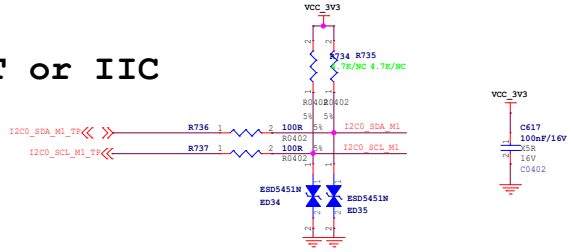
KEY



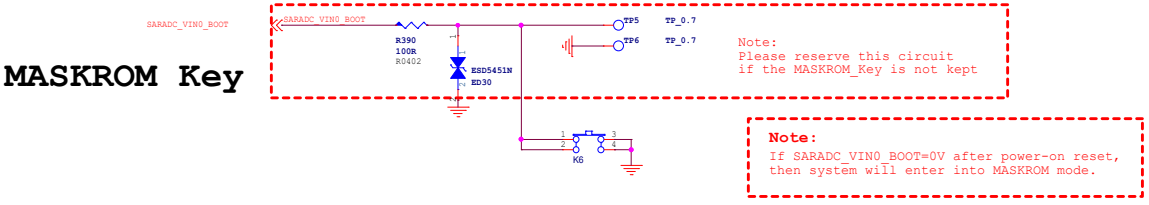
SARADC PIN

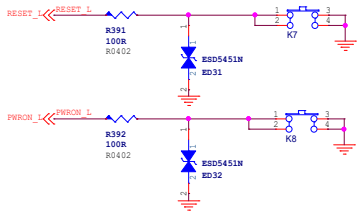


UART or IIC



MASKROM Key



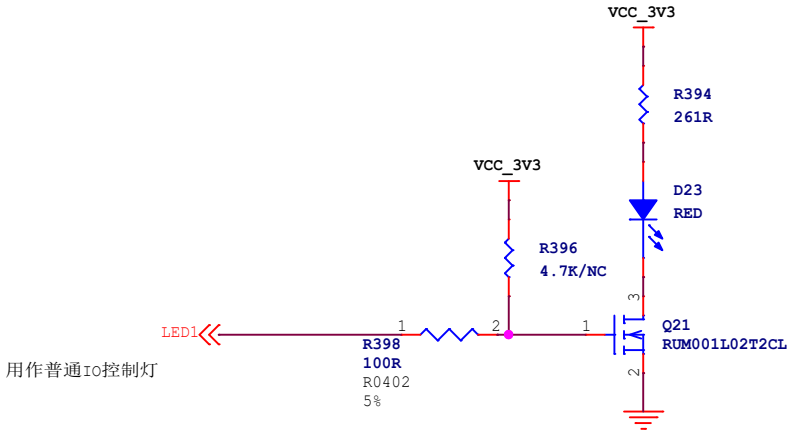


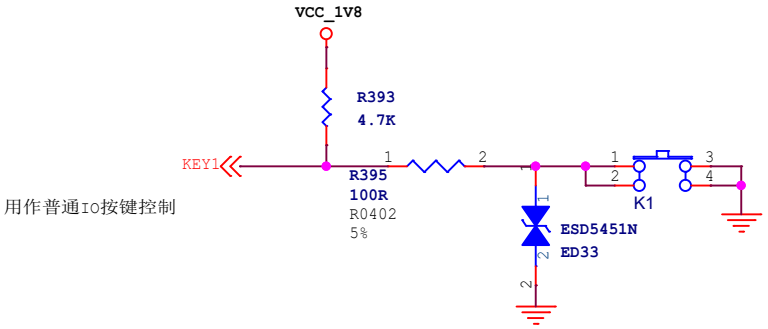
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Size: C	Page Title: 19.Maskrom/Reset/Pwr	Rev:	<RevCode>
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LED/KEY

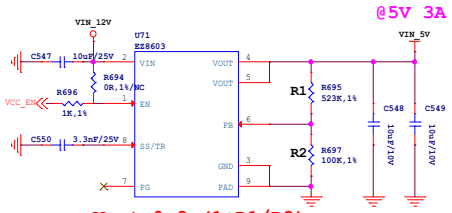




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DC/DC 5.0V

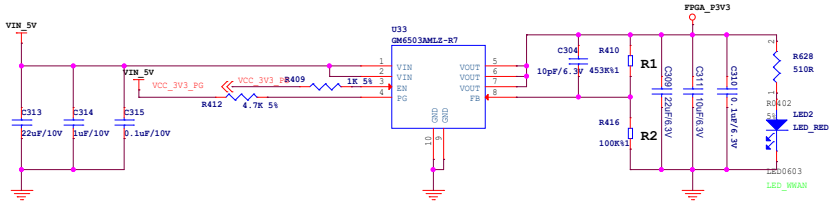
@5V 3A



Vout=0.8x(1+R1/R2)

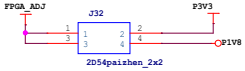
DC/DC 3.3V

@3.3V 3A



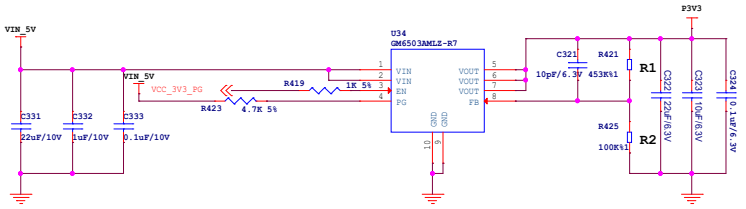
输入电压：2.5V-5.5V
输出电压：0.6V-VIN
输出电流：3A
工作温度范围：-40~125℃
VREF:0.6V±0.1%
VEN>1.1V

Vout=0.6x(1+R1/R2)



DC/DC 3.3V

@3.3V 3A

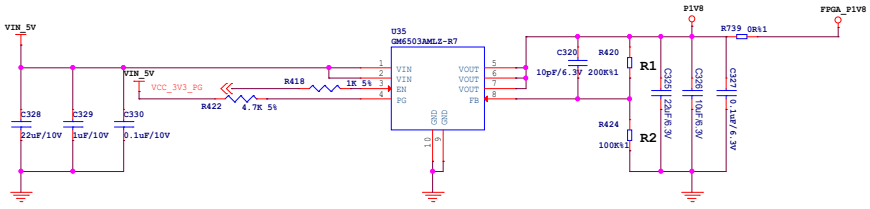


输入电压：2.5V-5.5V
输出电压：0.6V-VIN
输出电流：3A
工作温度范围：-40~125℃
VREF:0.6V±0.1%
VEN>1.1V

Vout=0.6x(1+R1/R2)

DC/DC 1.8V

@1.8V 3A



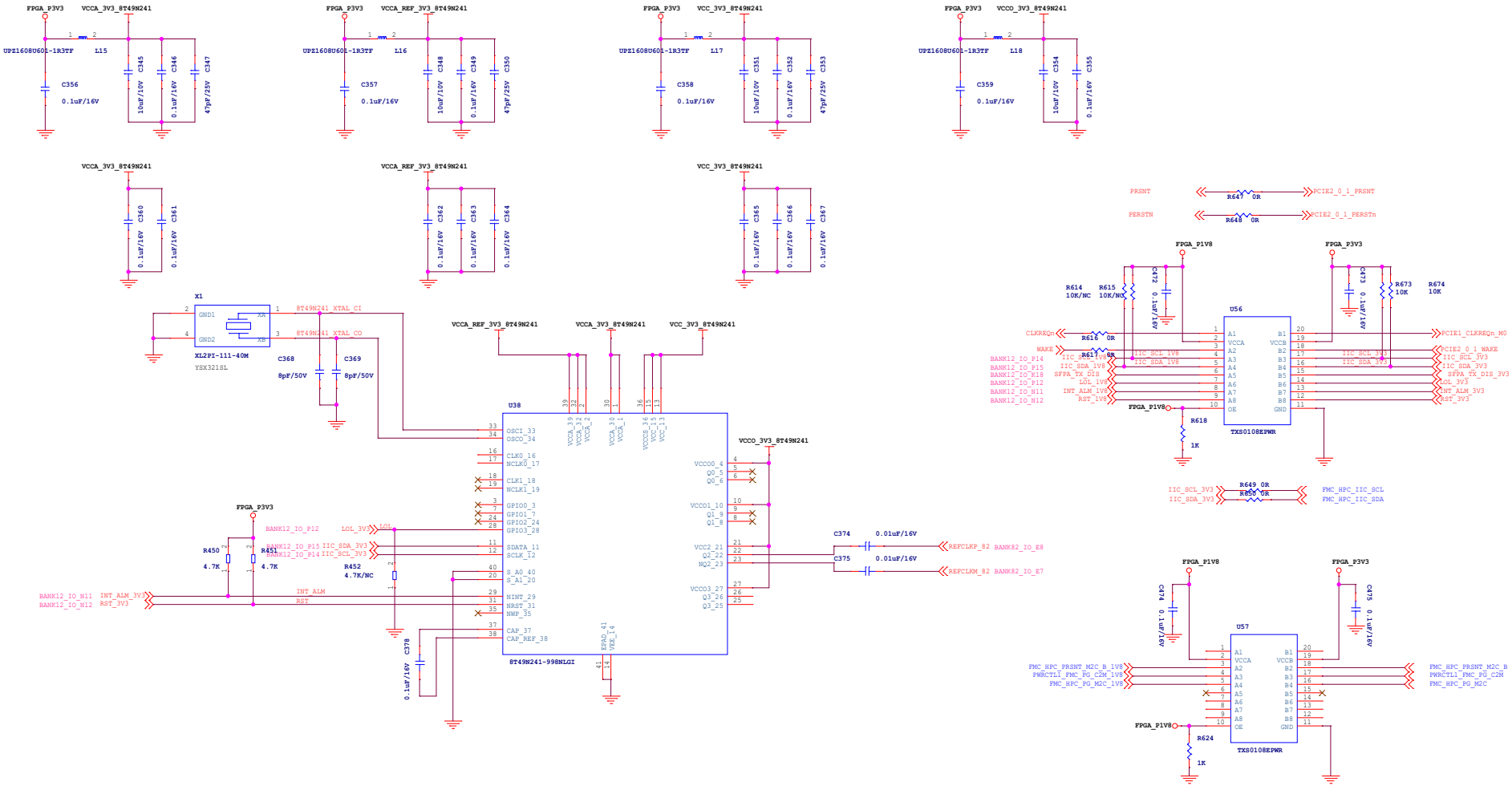
输入电压：2.5V-5.5V
输出电压：0.6V-VIN
输出电流：3A
工作温度范围：-40~125℃
VREF:0.6V±0.1%
VEN>1.1V

Vout=0.6x(1+R1/R2)

RK3576核心电源上电后----->VIN_5V

FPGA核心板电源使能------>FPGA_P3V3 / P3v3 / P1V8

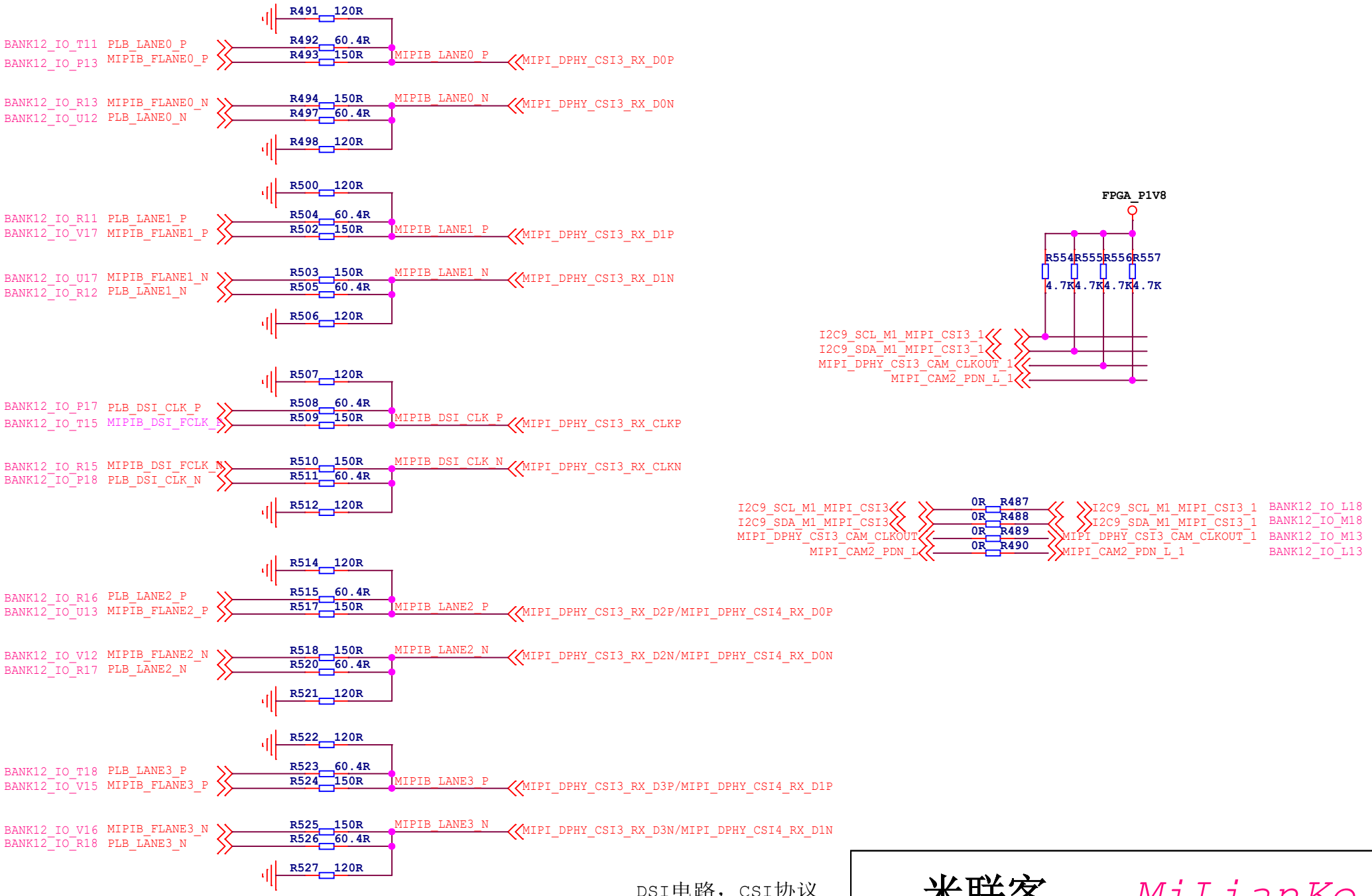
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ProjectName: MLK-AFH03
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ProjectName: MLK-AFH03			
Size: C	Page Title: 22.FPGA CLK	Rev:	<RevCode>
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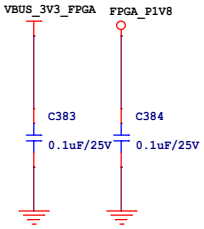
DSI电路，CSI协议

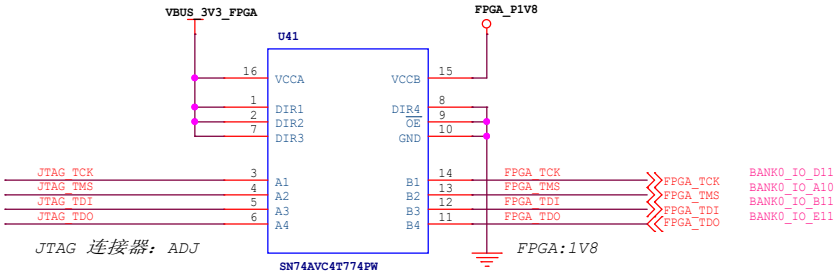
米联客 MiLianKe

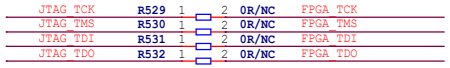
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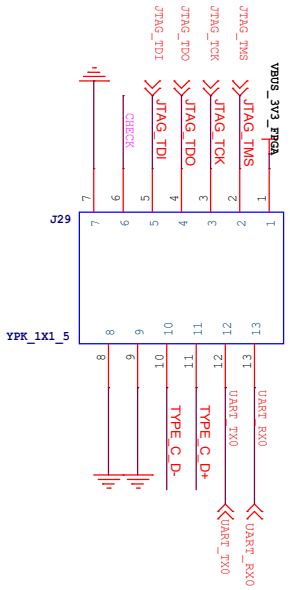
ProjectName: MLK-AFH03			
Size: A	Page Title: 23.FPGA_MIPI_DSI	Rev:	<RevCode>
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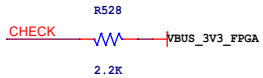
JTAG&UART

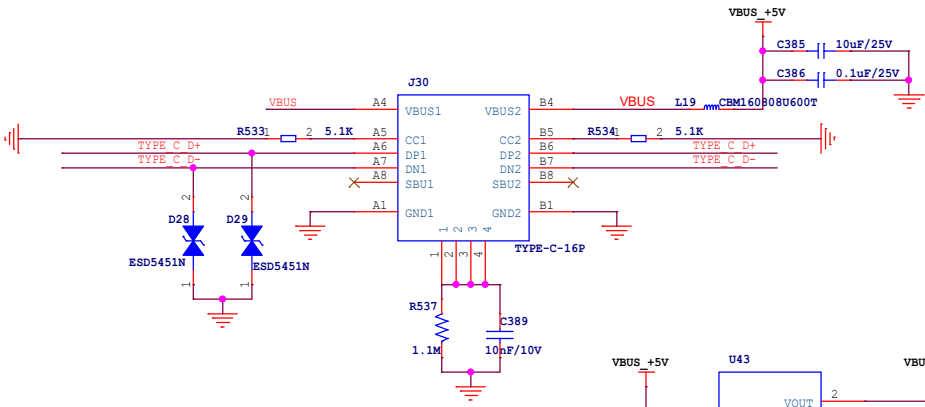


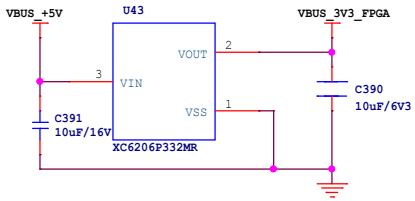




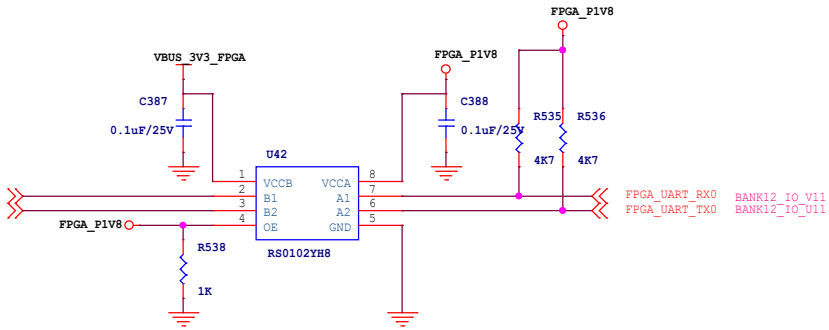








最大输入电压：8V
输出电压：3.3V
输出电流：250mA

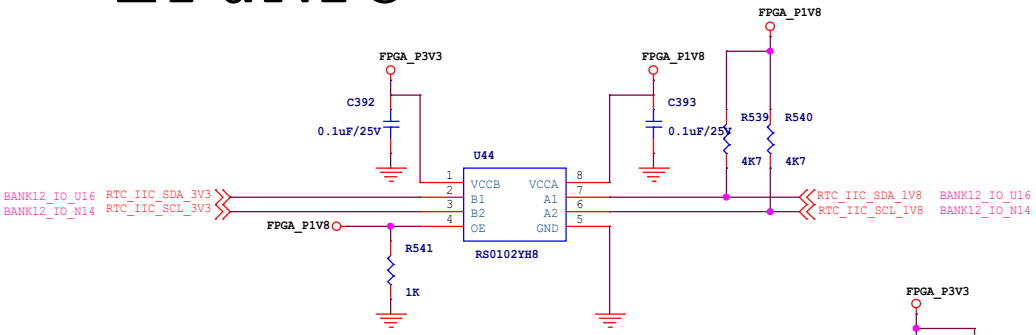


A端口为1.65V至5.5V
B端口为2.3V至5.5V
（VCCA≤VCCB）
-40°C至+85°C

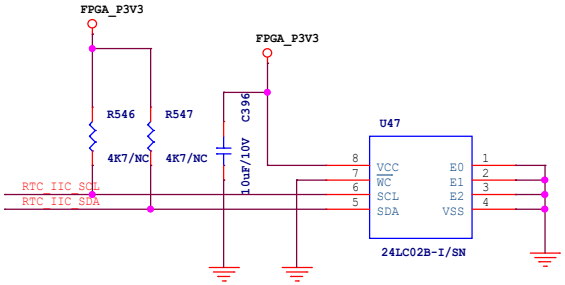
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ProjectName: MLK-AFH03			
Size: B	Page Title: 24.FPGA JTAG&UART	Rev:	<RevCode>
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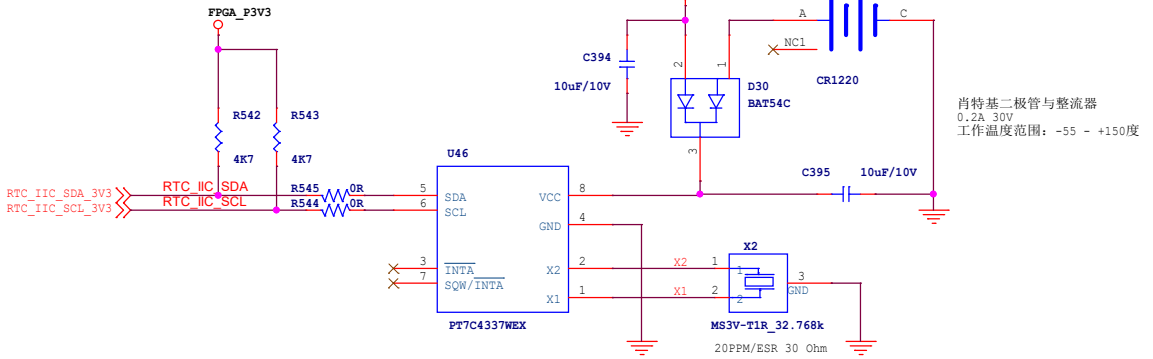
EP&RTC



A端口为1.65V至5.5V
B端口为2.3V至5.5V
（VCCA≤VCCB）
-40°C至+85°C



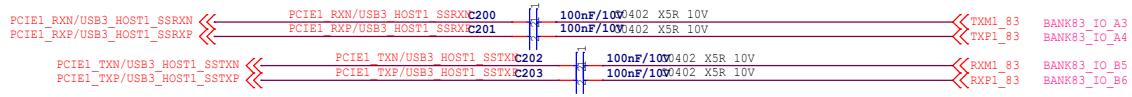
工作电压： 2.5V - 5.5V

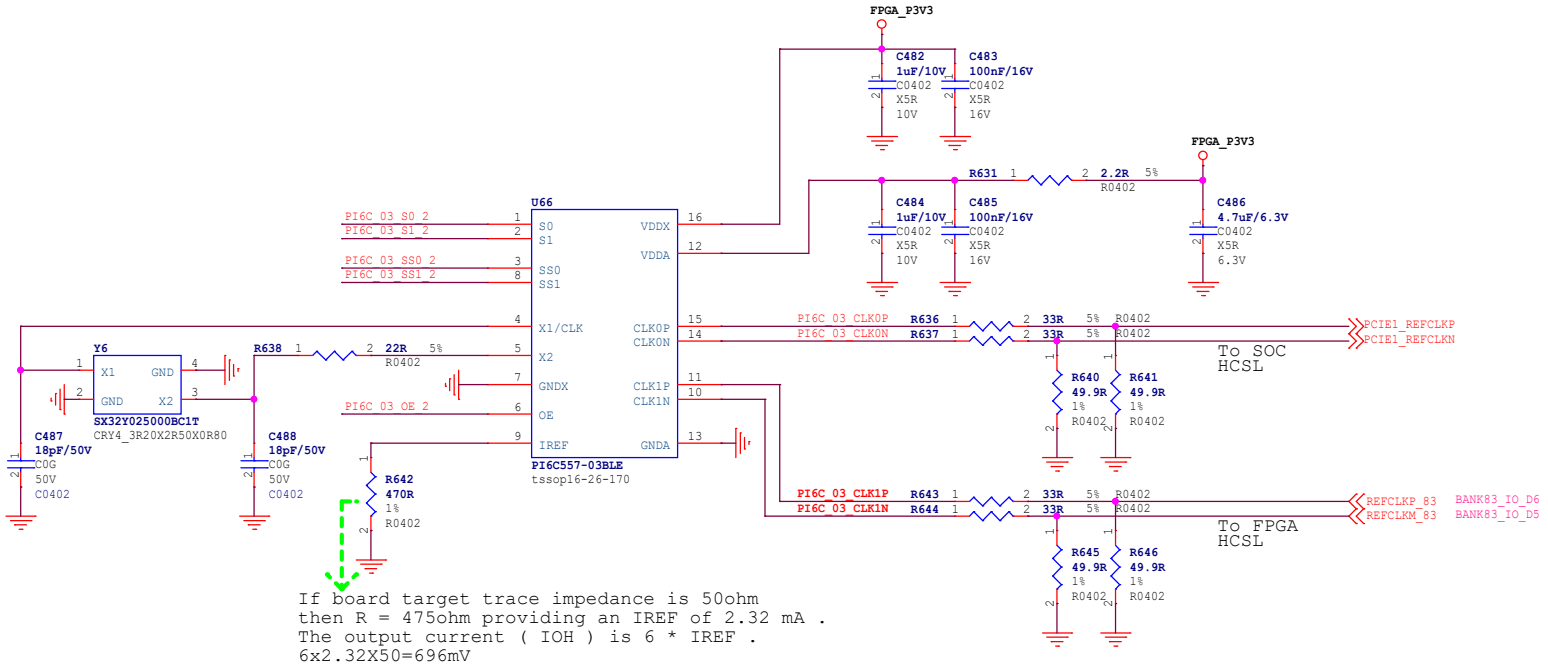


RTC时钟
工作温度范围：-40 - +85度
电源电压： 1.8V - 5.5V

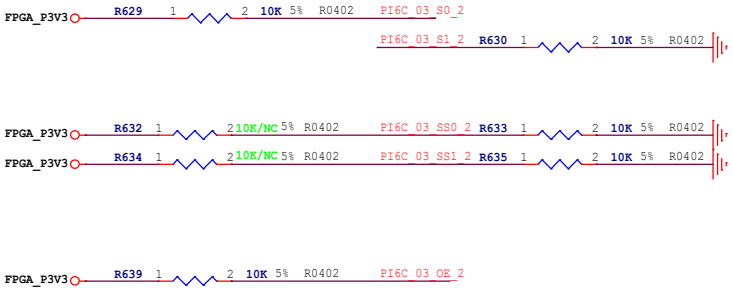
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ProjectName: MLK-AFH03			
Size: B	Page Title: 25.FPGA EP RTC	Rev:	<RevCode>
Date: Saturday, November 29, 2025	Sheet 26 of 29		





If board target trace impedance is 50ohm
then R = 475ohm providing an IREF of 2.32 mA .
The output current (IOH) is 6 * IREF .
6x2.32X50=696mV

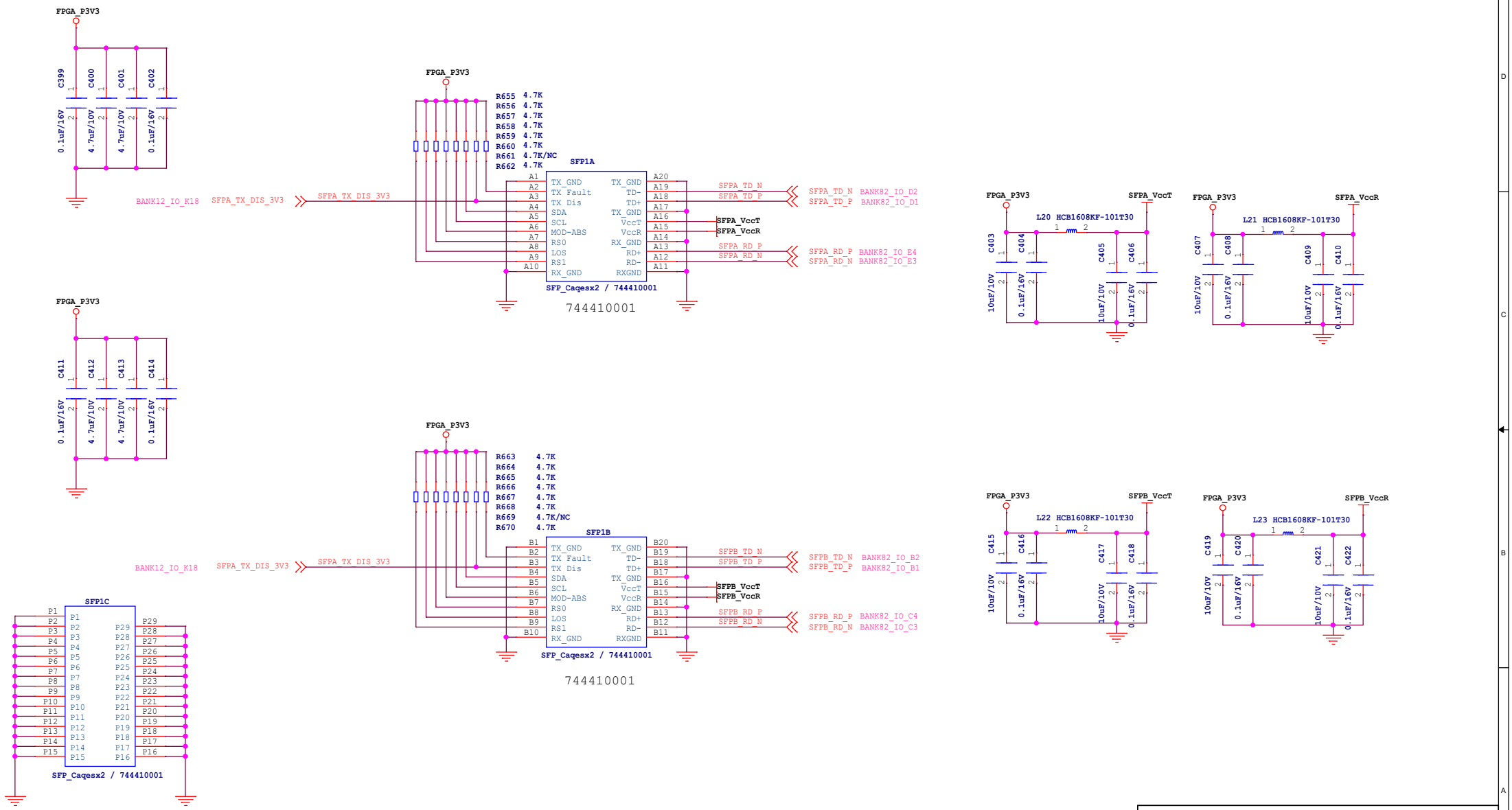


PI6C_S1	PI6C_S0	Out Freq
0	1	100MHz

PI6C_SS1	PI6C_SS0	Spread %
0	0	No Spread
0	1	-0.5
1	0	-1.0
1	1	No Spread

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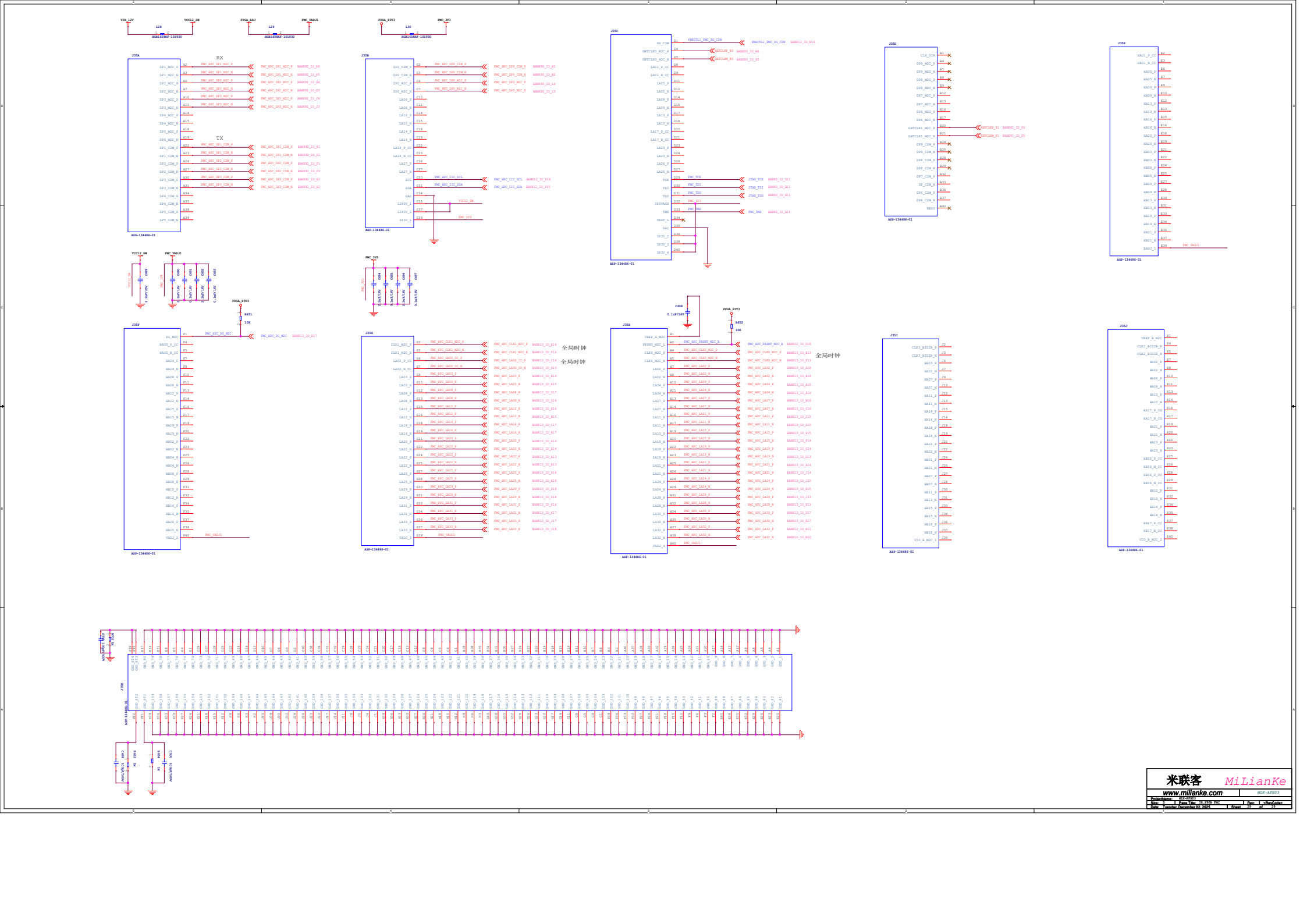
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FPGA_P3V3
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